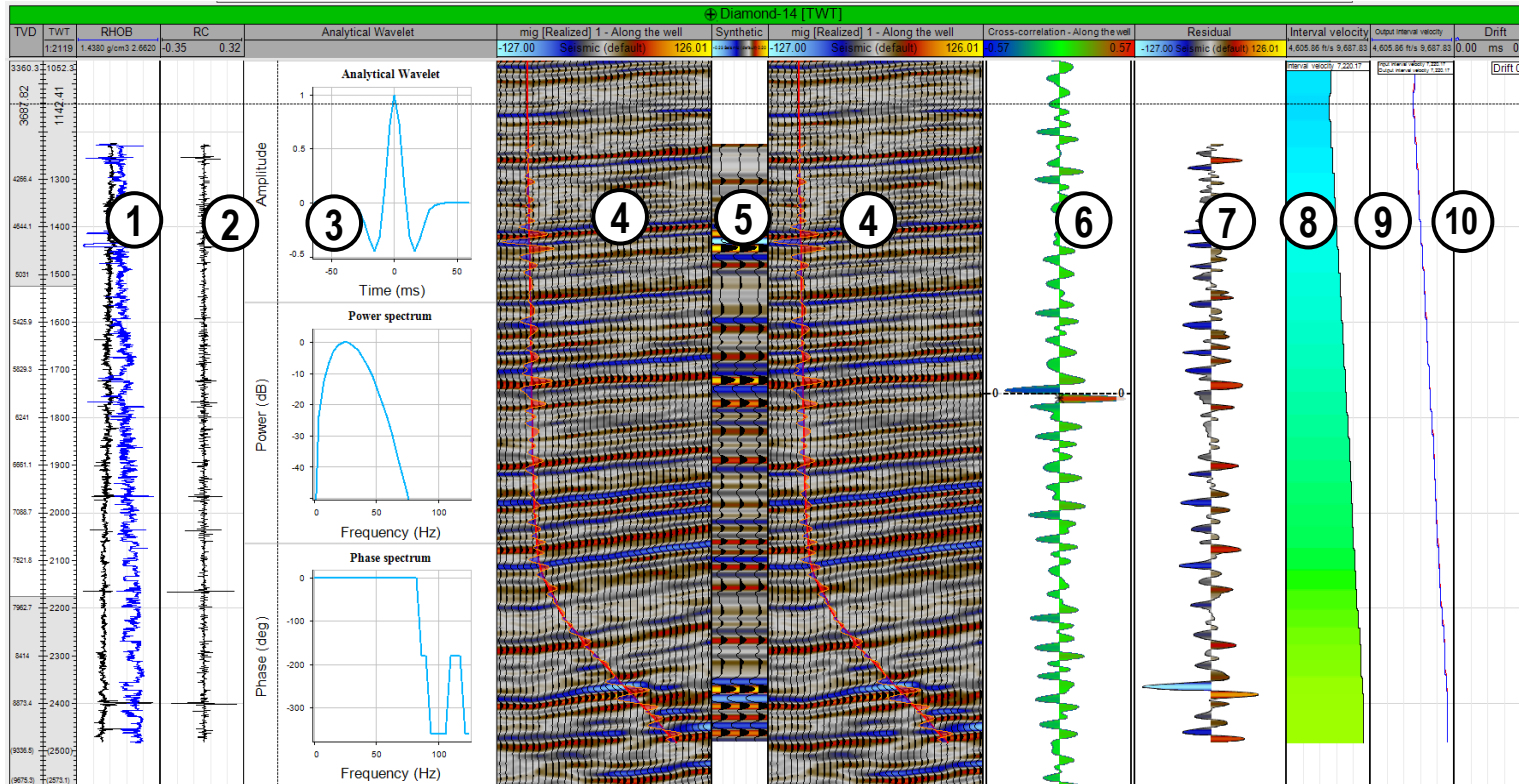


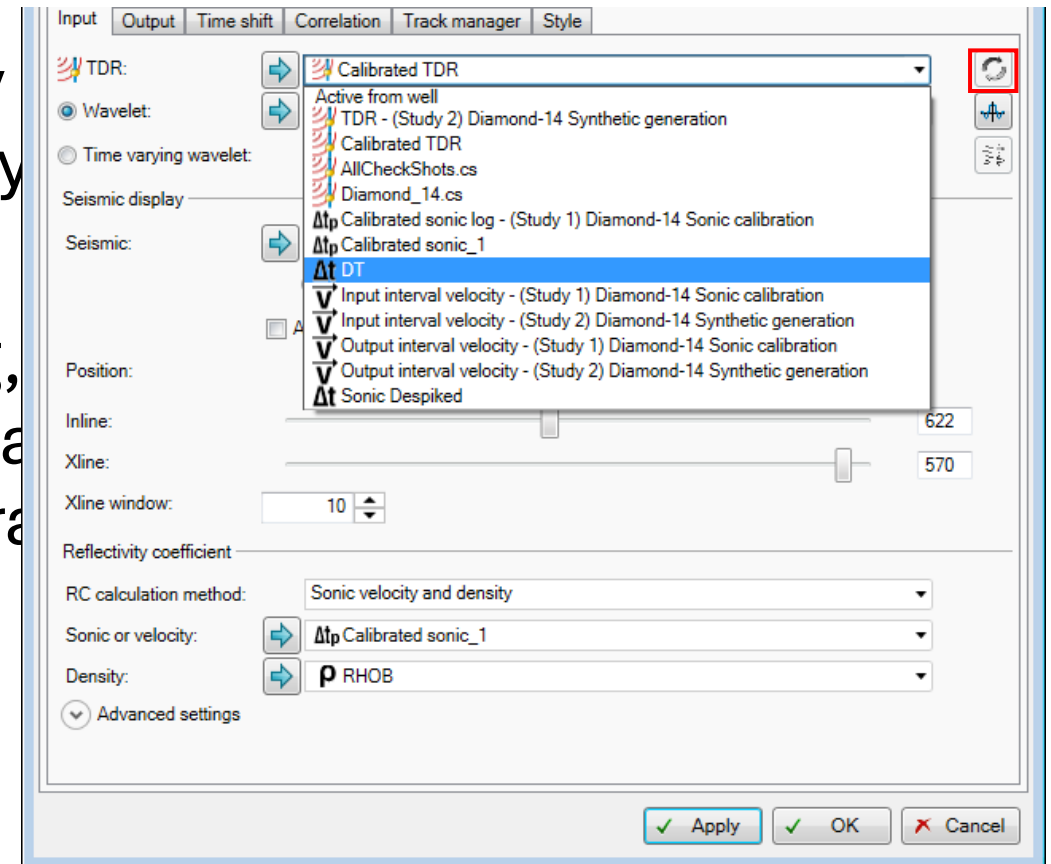
Generate a synthetic seismogram



- 1 Input logs track
- 2 Reflectivity track
- 3 Wavelet track
- 4 Seismic track
- 5 Synthetic track
- 6 Cross-correlation track
- 7 Residual track
- 8 Interval Velocity track
- 9 Input/Output Interval Velocity track
- 10 Drift track

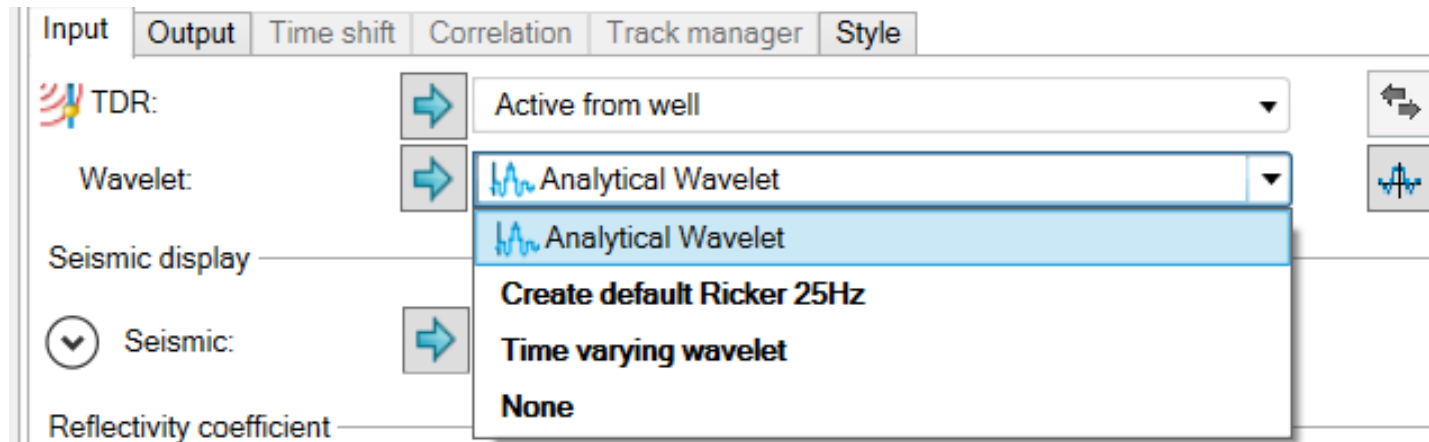
Select TDR input

- Seismic to well tie process is very interactive. You can select many types of TDR as input in the study.
- You can select any TDR, sonic log, velocity log contained by the well as TDR input for the synthetics generation study.



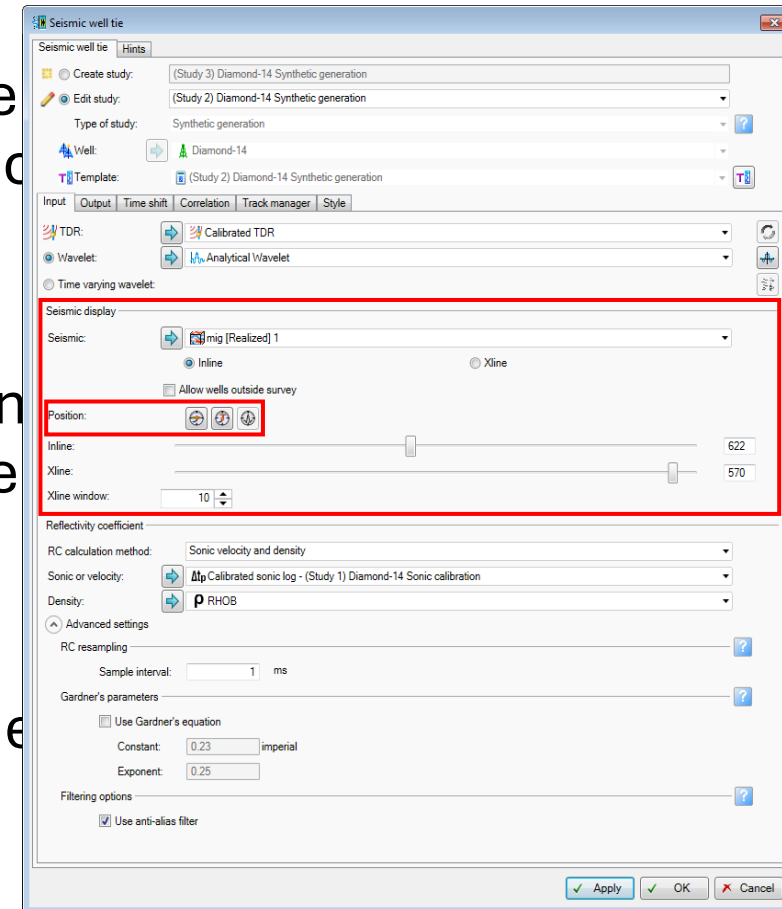
Wavelet

- Insert the wavelet to be convolved with the reflectivity series to create a synthetic seismogram.
- If no wavelet is available in the project, open the wavelet toolbox to create a new one by clicking *Lau*  *wavelet toolbox* .



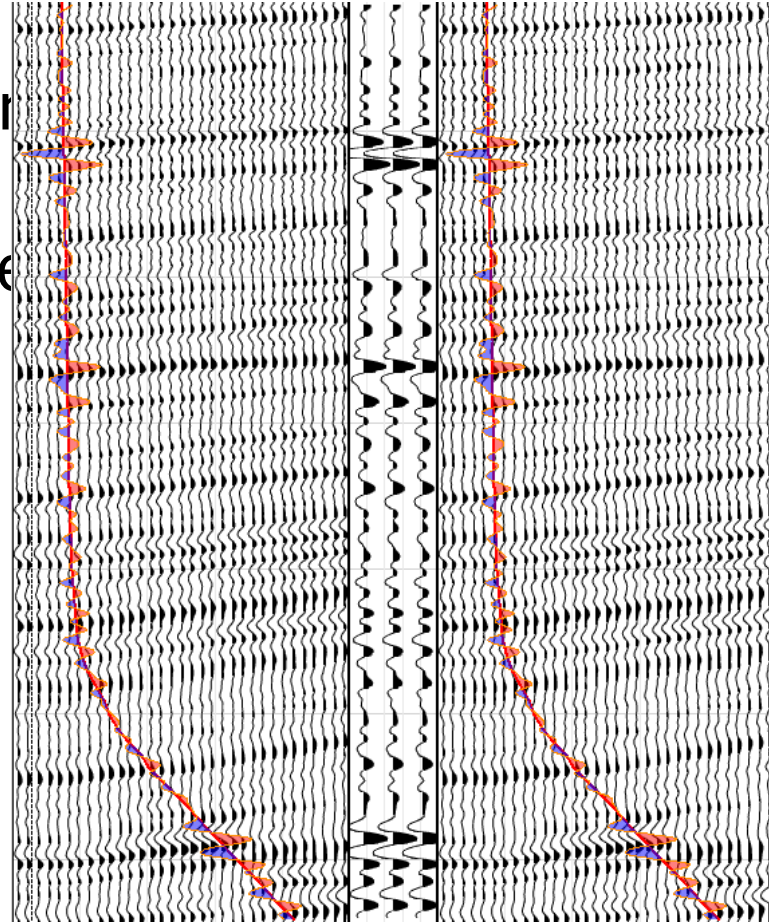
Seismic display

- In the **Seismic display** section, you insert seismic that is used as a reference to compare the seismic-synthetic tie.
- You can use a 3D cube or a 2D line.
- There are four seismic display position methodologies for the Seismic well tie
 - Seismic along the well trajectory
 - Wellhead location on Inline/Xline
 - Deviated well location on Inline/Xline
 - Selected Wavelet on Inline/Xline



Seismic along the well trajectory option

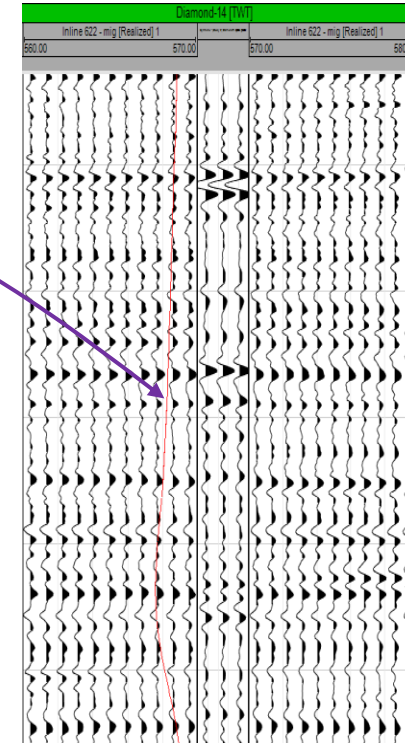
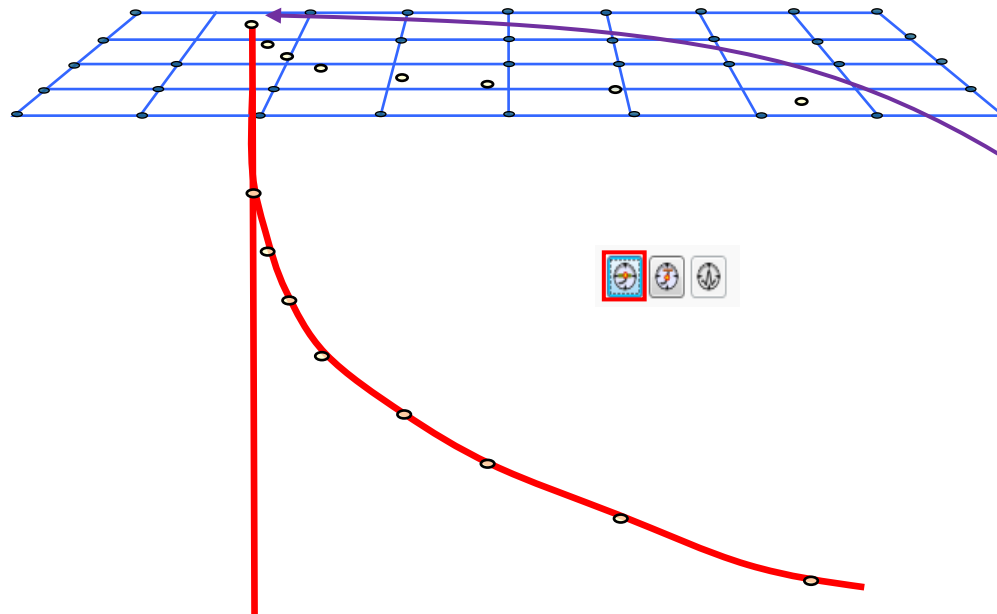
- This option is the default visualization
- The seismic is displayed along the well trajectory. The option allows a deviated well.



Wellhead location

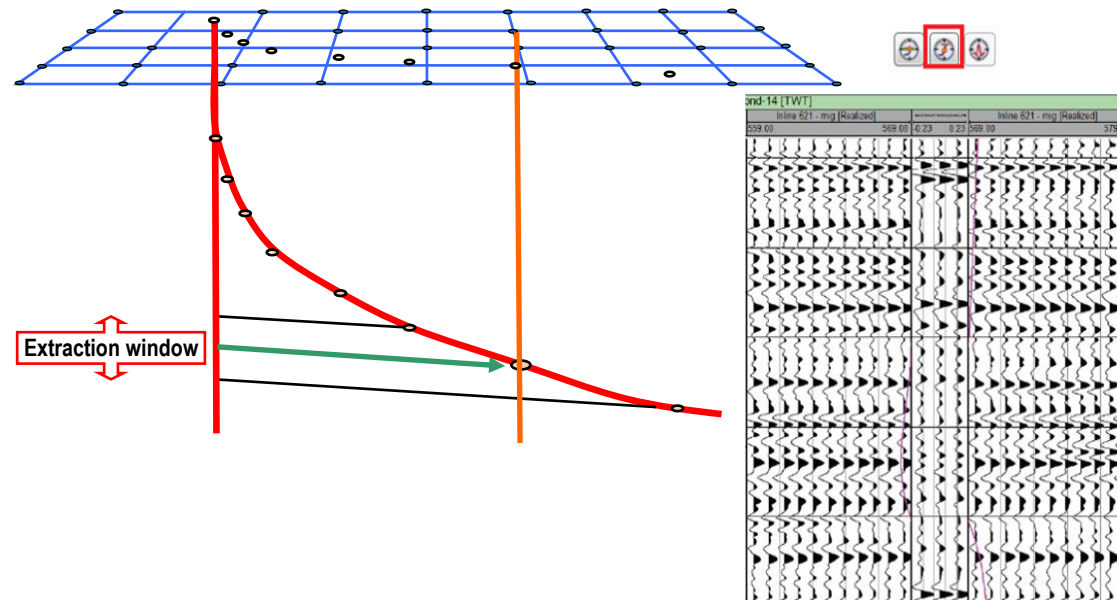
- Set the Inline-Xline to the location of the wellhead.

Position is set at the well surface XY location.



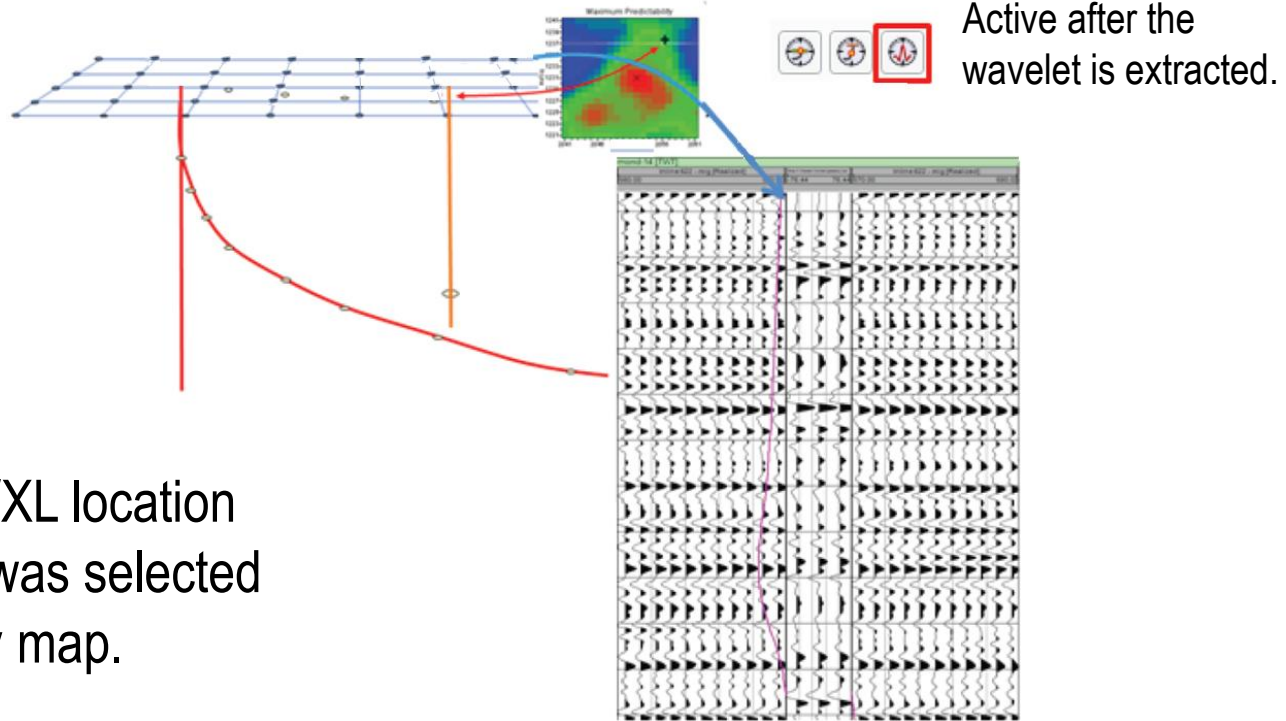
Deviated well location

- Set the Inline-Xline to the location of the well track at the center of the analysis window. Position is set at the XY location where the trajectory crosses time at the center of the extraction window.



Selected wavelet

- Set the Inline-Xline to the location of the currently selected wavelet.



Position is set at IL/XL location where the wavelet was selected on the Predictability map.

Reflectivity coefficient calculation

- Depending on the availability of data, the Reflectivity coefficient can be calculated using these methods:

- Impedance (AI/ EI/ EEI)
- Sonic and density logs
- Shear
- Porosity
- Aki and Richards PP
- Aki and Richards PS
- Any log

The screenshot shows the 'Seismic well tie' dialog box with the 'Hints' tab selected. The 'Create study' section is set to 'Synthetic generation'. The 'Well' is 'Diamond-14'. The 'Template' is 'Synthetic generation default template'. The 'Reflectivity coefficient' section is highlighted with a red box and contains the following settings:

- RC calculation method: Sonic velocity and density
- Sonic or velocity: Δt DT
- Density: ρ DRHO
- Advanced settings:
 - RC resampling: Sample interval: 1 ms
 - Sonic/density computation parameters:
 - ☒ Use sonic/density computation
 - ☒ Gardner's equation: Constant: 0.23, Exponent: 0.25 imperial
 - ☐ Constant value: Sonic: 30.48 us/ft, Density: 2.65 g/cm3
 - RC options:
 - ☒ Primary only, ☐ Multiples only, ☐ Primary & Multiples
 - Surface Reflection Coefficient [0.0 - 1.0]: 1
 - Time shift correction:
 - ☒ Plus half, ☐ Zero, ☐ Minus half
 - Filtering options:
 - ☒ Use anti-alias filter

Buttons at the bottom: Apply, OK, Cancel.

Gardner's patching: Auto-complete reflection coefficient inputs

- On-the-fly use of Gardner's equation to compute DT or Density log during RC calculation.
- The patch shows that RC and Synthetic can be calculated for a missing section of the Sonic using the *Gardner's equation* option.

Advanced settings

RC resampling

Sample interval: ms

Sonic/density computation parameters

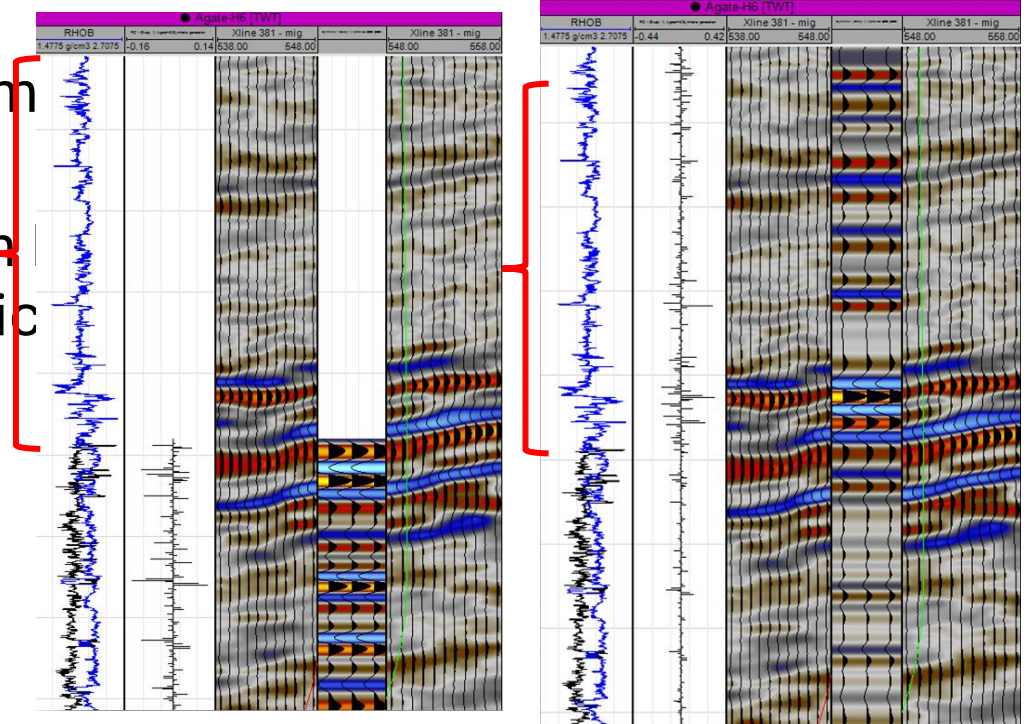
☒ Use sonic/density computation

☒ Gardner's equation Constant:

Exponent: imperial

☐ Constant value Sonic: us/ft

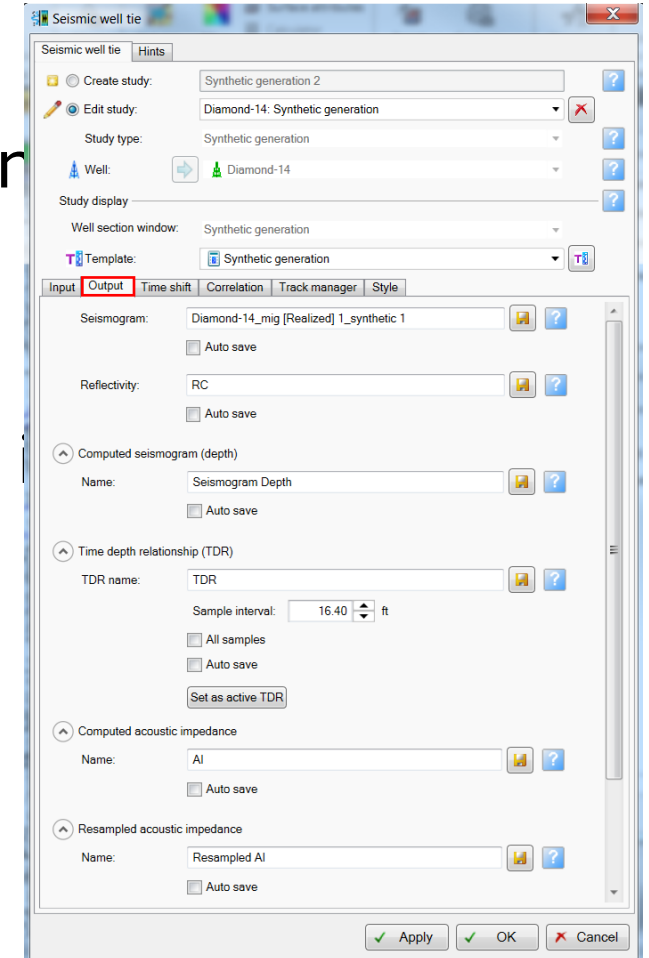
Density: g/cm3



Output tab

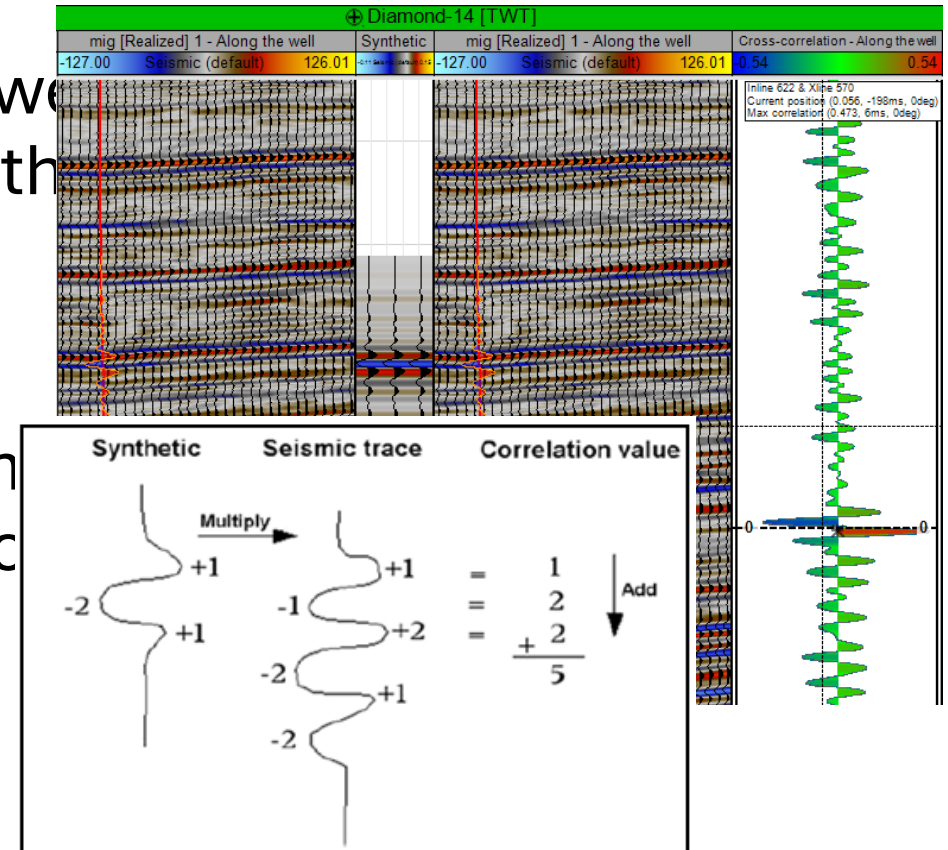
■ Use the **Output** tab in the Synthetic generation control the output results from the workflow.

■ To change the name of any output and save it in the **Input** pane, click  Save .



Correlation track (1)

- Calculates the cross-correlation between synthetic and seismic by calculating the shift to be applied to the synthetic.
- This correlation value is the sum of the products of the synthetic and seismic amplitudes at each position in time.



Correlation track (2)

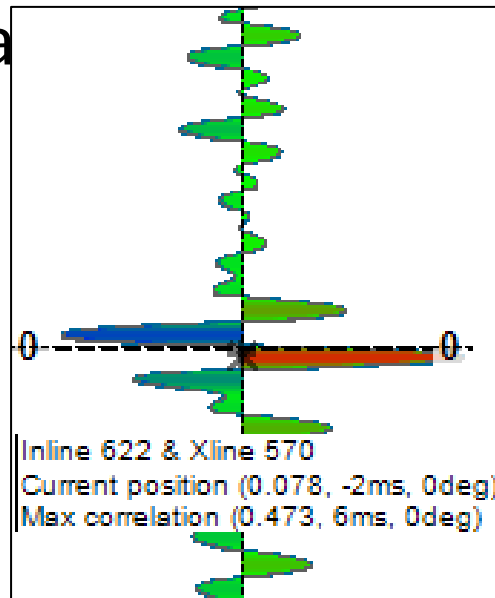
- When you place the cursor over the Correlation track, you can read the inline and crossline positions, the correlation value that corresponds with the mouse position (Current position), and the Max correlation from the box labeled in the upper right side of the track

Inline 622 & Xline 570
Current position (0.033, -200ms, 0deg)
Max correlation (0.473, 6ms, 0deg)

- The lag value at any point in the Correlation track is the time shift applied to the synthetic to move it into the position where it was when the correlation values were calculated.

Correlation track (3)

A dotted line marks the trace posted on the Correlation track with the point of highest correlation. The lag value at this point is the amount of bulk shift that you would need to apply to move the synthetic into this best match.



Correlation tab (1)

This tab controls the display parameters of the Correlation track.

It has three main sections:

- Window
- Trace
- Phase mistie

The screenshot shows the 'Correlation' tab of a software interface. At the top, there are five tabs: 'Input', 'Output', 'Time shift', 'Correlation' (which is active), 'Track manager', and 'Style'. The main area is divided into three sections, each with a title bar and a help icon (a blue square with a white question mark):

- Window**: Contains a checked checkbox for 'Auto set limits'. Below it are two input fields: 'Start time:' with the value '0' and 'ms', and 'End time:' with the value '3500' and 'ms'. At the bottom of this section is an unchecked checkbox for 'Apply same limits to correlation'.
- Trace**: Contains three radio button options: 'Match seismic traces' (which is selected), 'User defined', and 'Single trace'. Below 'User defined' are two input fields: 'Start trace:' with the value '9' and 'End trace:' with the value '9'. Below 'Single trace' is an input field for 'Radius:' with the value '1'.
- Phase mistie**: Contains an unchecked checkbox for 'Compute phase mistie'.

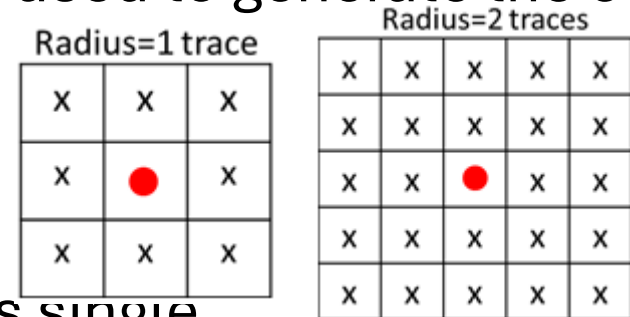
Correlation tab (2)

- **Match seismic trace:** Visualize the generated correlation traces using the entire range of the reference seismic, defined in the **Seismic display** section of the **Seismic well tie** dialog box.

User defined: Specify the range of traces to be used to generate the correlation.

Single trace:

- A single trace is calculated following the well trajectory.
- The traces used in the calculation for this single trace (average trace) are the traces contained within a radius that you determine.



● Represents the well trajectory at time i
x Represents the traces around the well trajectory

Correlation tab (3)

Phase mistie

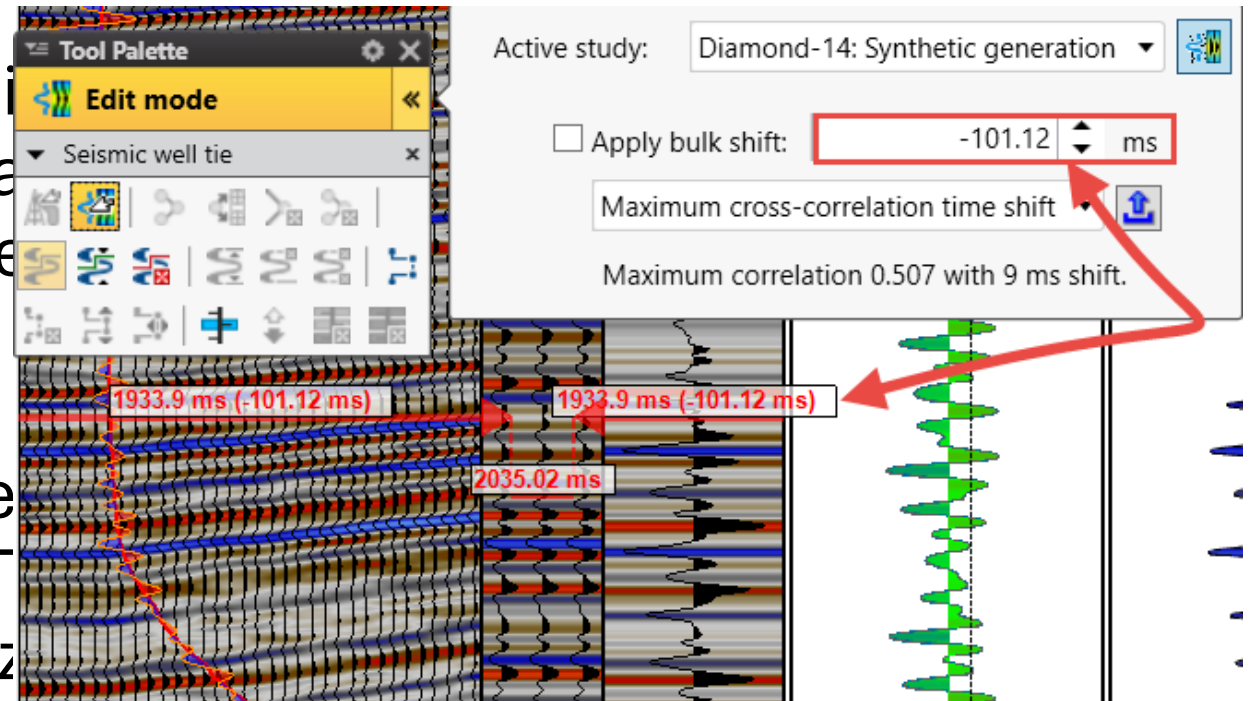
- Calculates the correlation between synthetic/seismic while considering phase of the wavelet.

The screenshot displays the 'Correlation' tab of a software interface. The tab is highlighted with a red border. The interface is divided into three main sections: 'Window', 'Trace', and 'Phase mistie'. Each section has a title bar with a question mark icon.

- Window:** Contains a checked checkbox for 'Auto set limits'. Below it are input fields for 'Start time: 0 ms' and 'End time: 3500 ms'. A checkbox for 'Apply same limits to correlation' is unchecked.
- Trace:** Contains three radio button options: 'Match seismic traces' (selected), 'User defined', and 'Single trace'. Under 'User defined', there are input fields for 'Start trace: 0' and 'End trace: 14'. Under 'Single trace', there is an input field for 'Radius: 1'.
- Phase mistie:** Contains a checked checkbox for 'Compute phase mistie', which is highlighted with a red border.

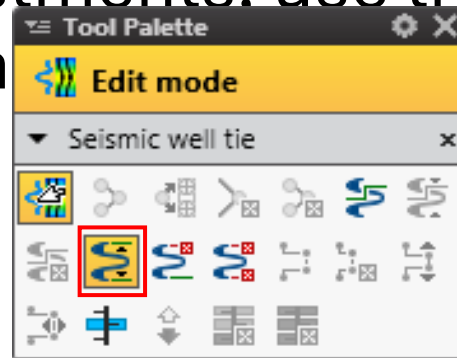
Use interactive bulk shift

- The next step is to do a bulk shift of the synthetic to get a closer match between the synthetics and the seismic.
-
- If the quality of the tie is good enough, a stretch and squeeze will NOT be necessary. (Stretch and squeeze is not recommended as the first approach.)

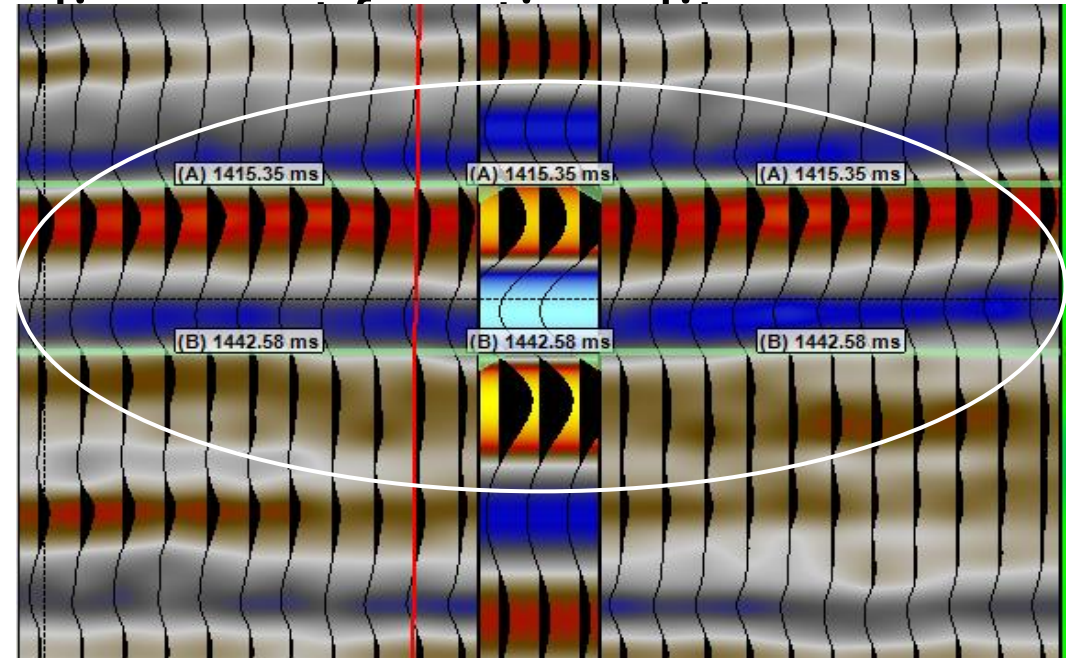


Use continuous alignment

- Sometimes, small adjustments must be made to match the synthetics with the seismic after the bulk shift is implemented. To perform these adjustments, use the continuous alignment tool. The process is interactive.

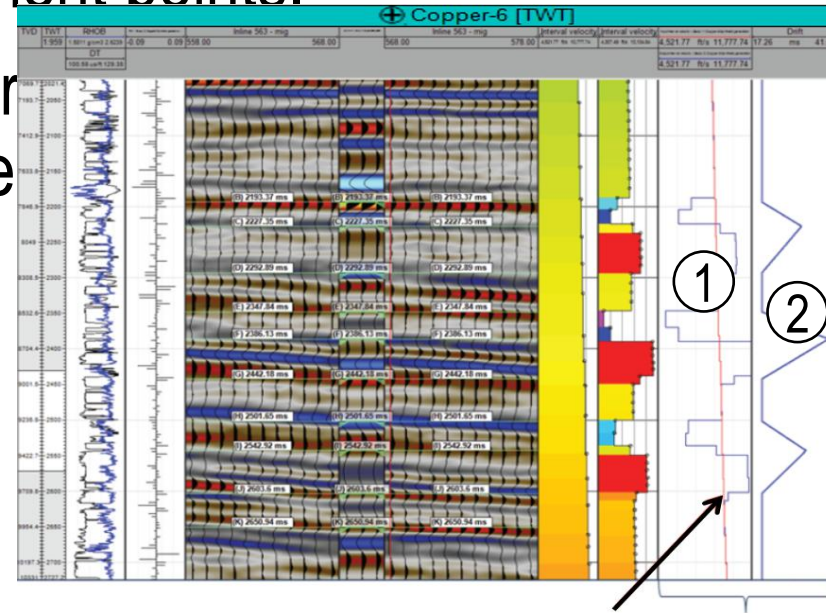


Note: You can add as many alignment points as required.



QC interval velocity

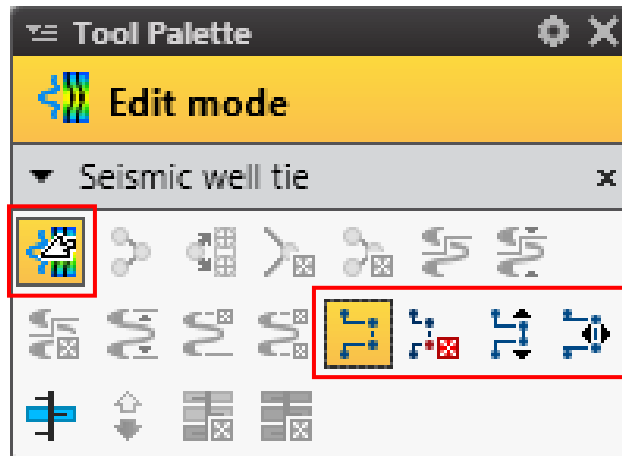
- The Input Interval Velocity vs. Output Interval Velocity and the Drift curve in time tracks show information about the resultant interval velocity and the time shift applied to the alignment points.
- These tracks act as guides for the synthetic generation process and velocity editing.



- 1 Input Interval Velocity vs. Output Interval
- 2 Drift Curve in time for shift applied

Interval velocity manipulation (1)

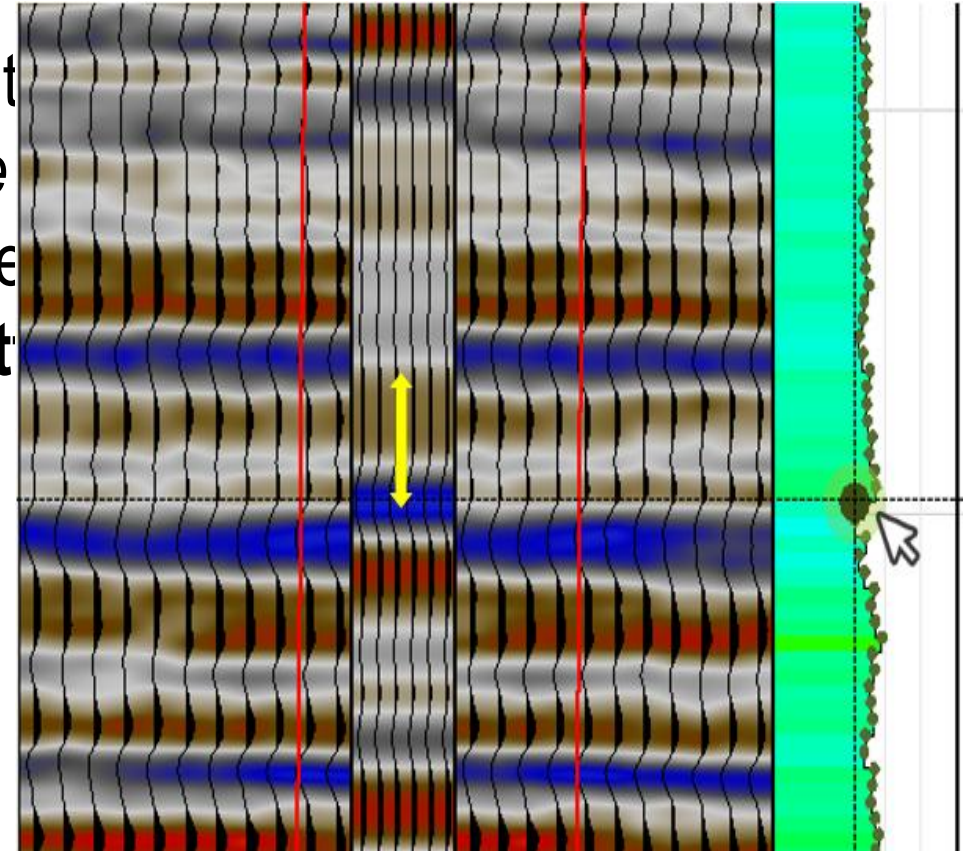
- Edit/manipulate the interval velocity on the fly to check the impact on the data.
- Edit the velocity values visually in a **Well section window**.
- Use the **Seismic well tie Tool Palette** to manipulate and delete velocity points.
- Lock the velocity manipulation into vertical and horizontal directions.



-  Manipulate interval velocity
-  Delete interval velocity knee
-  Vertical interval velocity manipulation
-  Horizontal interval velocity manipulation
-  Edit mode

Interval velocity manipulation (2)

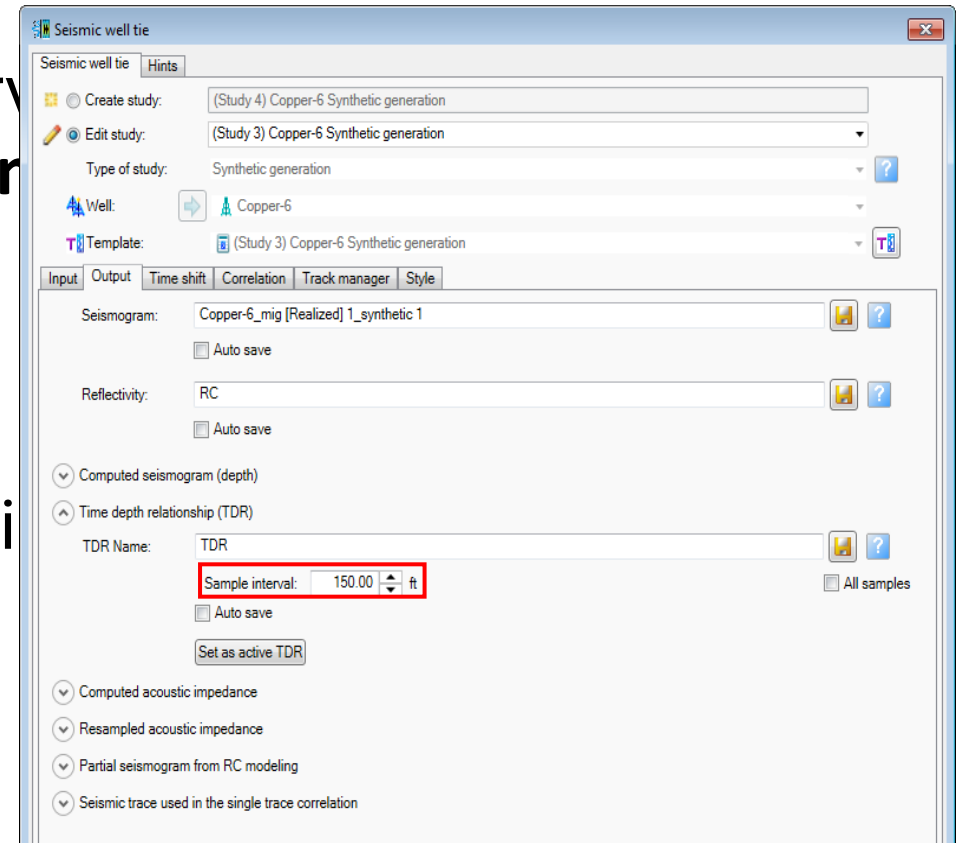
- In the Interval velocity track, you can manipulate the points when the Seismic well tie *Edit move* and *Manipulate interval velocity* buttons are switched on in the **Seismic well tie Tool Palette**



Interval velocity manipulation (3)

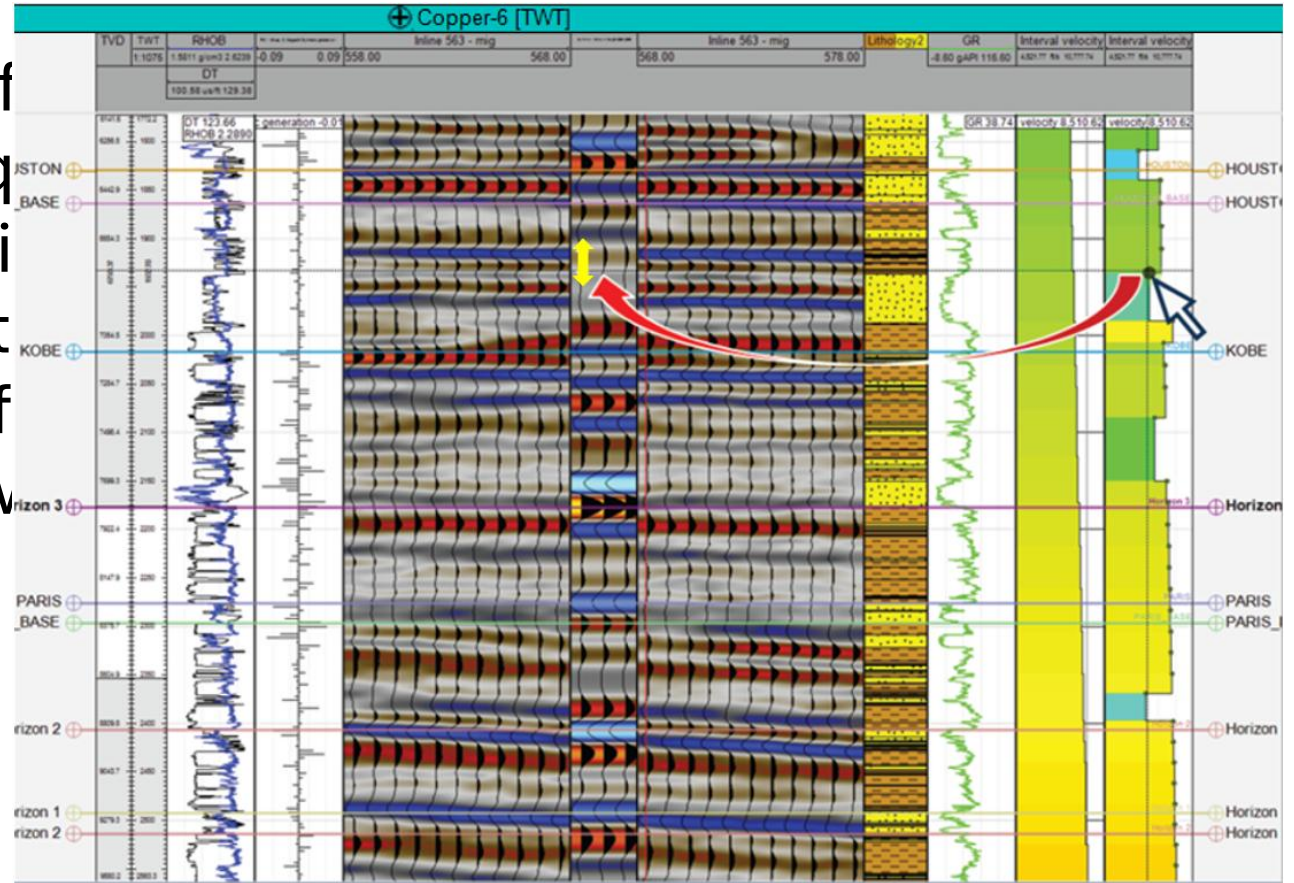
- You can set the sampling of the interval velocity on the **Output** tab of the **Seismic well tie** dialog box in the **Time/depth relationship (TDR)** section.

- Remember that the interval velocity is an attribute of the checkshot.



Interval velocity manipulation (4)

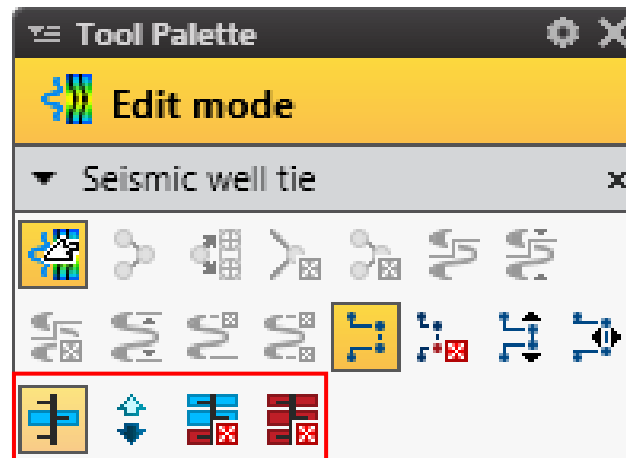
- The interactive manipulation of interval velocity can be an important tool for quality control purposes, especially if you have information about the lithology (from the driller or geologist, for example) because you know the range of velocity to expect.



Reflection coefficient (RC) modeling (1)

The RC modeling process allows you to:

- select/deselect reflection coefficients and see the impact in the new synthetic created through RC modeling
- study tuning effects
- filter problematic areas from your synthetic.



Activate RC modeling



Invert the RC selection



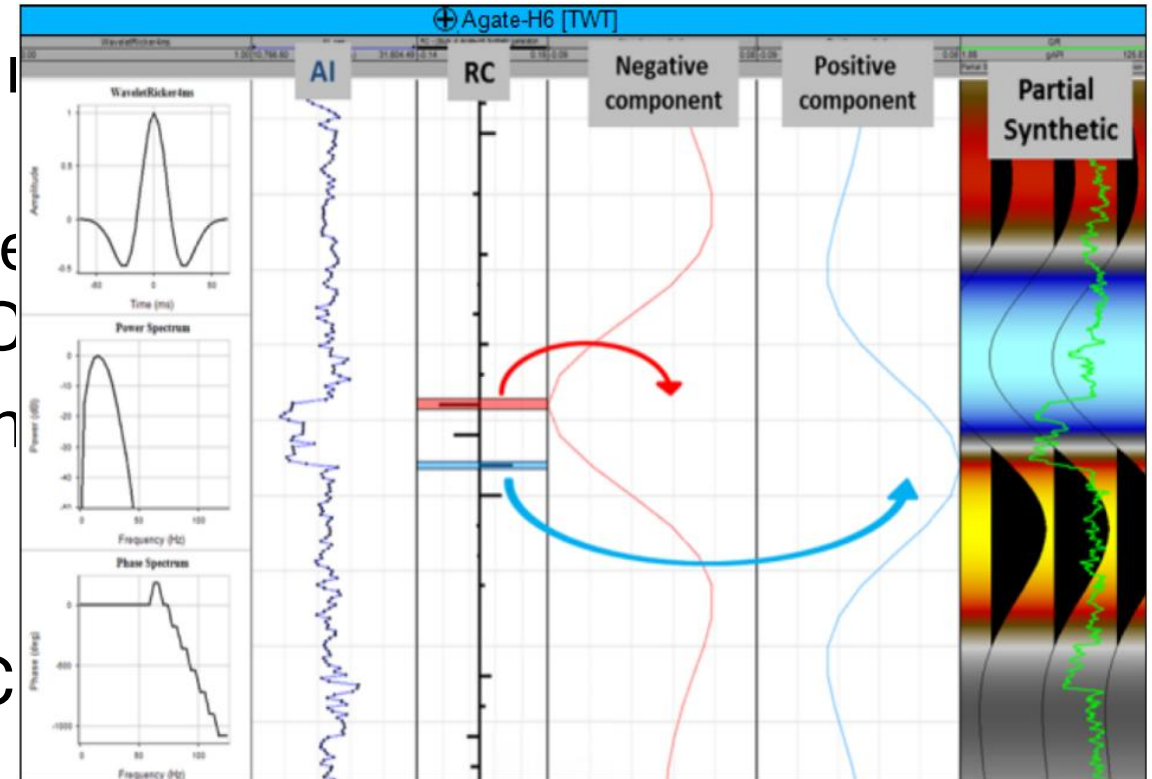
Delete single RC selection



Delete all RC selections

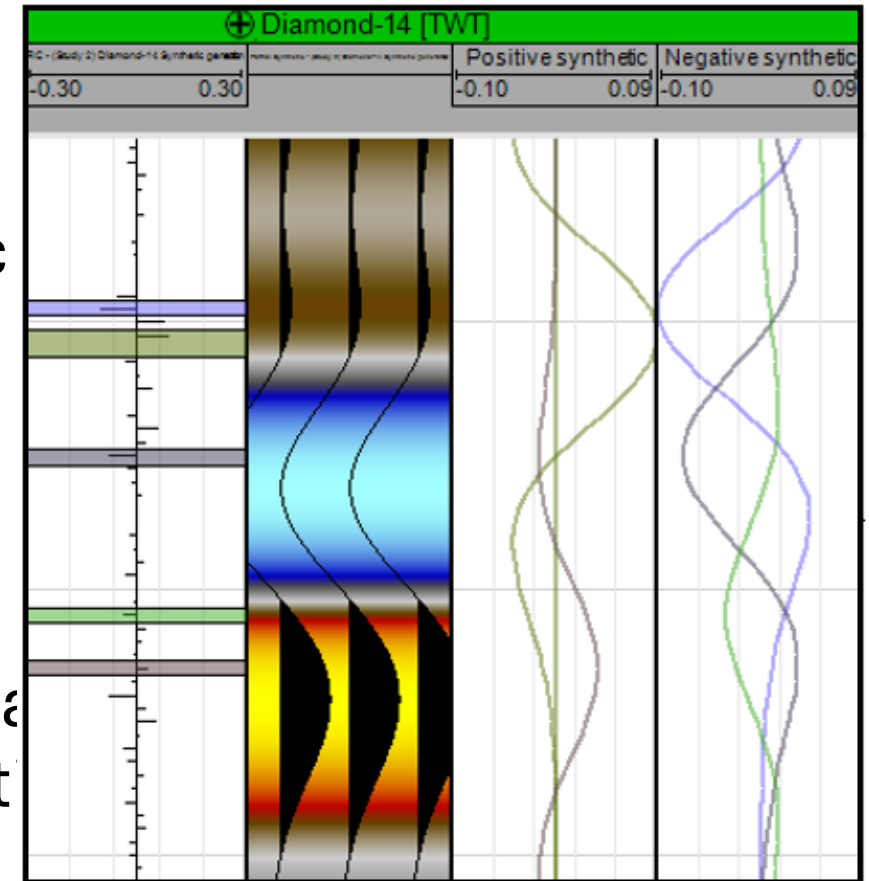
Reflection coefficient (RC) modeling (2)

- The result appears in three tracks in the **Well section window**:
 - Partial Synthetic (represents the synthetic from your selected RC)
 - Positive Synthetic (positive component of your selected RC values).
 - Negative Synthetic (negative component of your selected RC values)



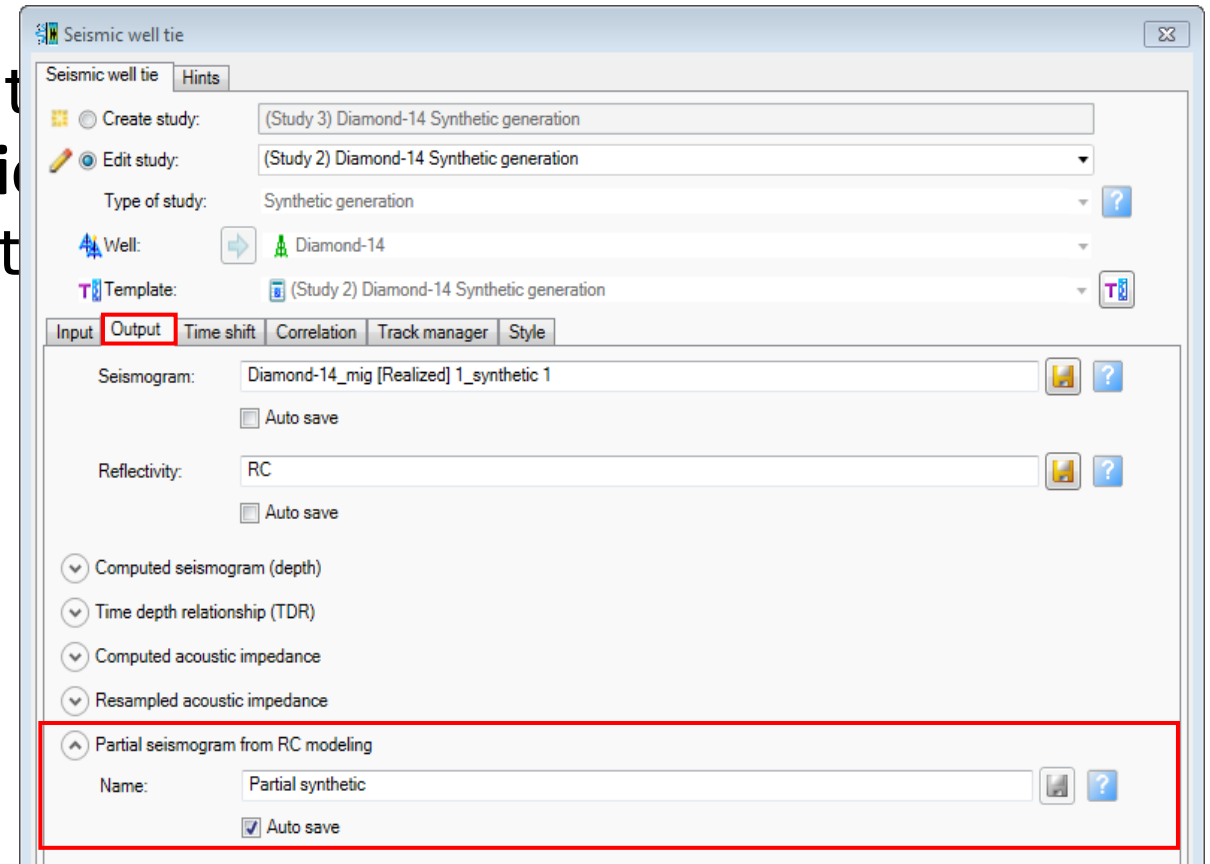
Run Reflection coefficient (RC) modeling (1)

1. In the **Seismic well tie Tool Palette**, click *Activate RC mode*  .
Three more tracks appear in the Synthetic of the **Well section window** (WSW):
 - Partial Synthetic
 - Positive component
 - Negative component
2. From the RC series track, select/clear area on your log to see the impact on your synthetic



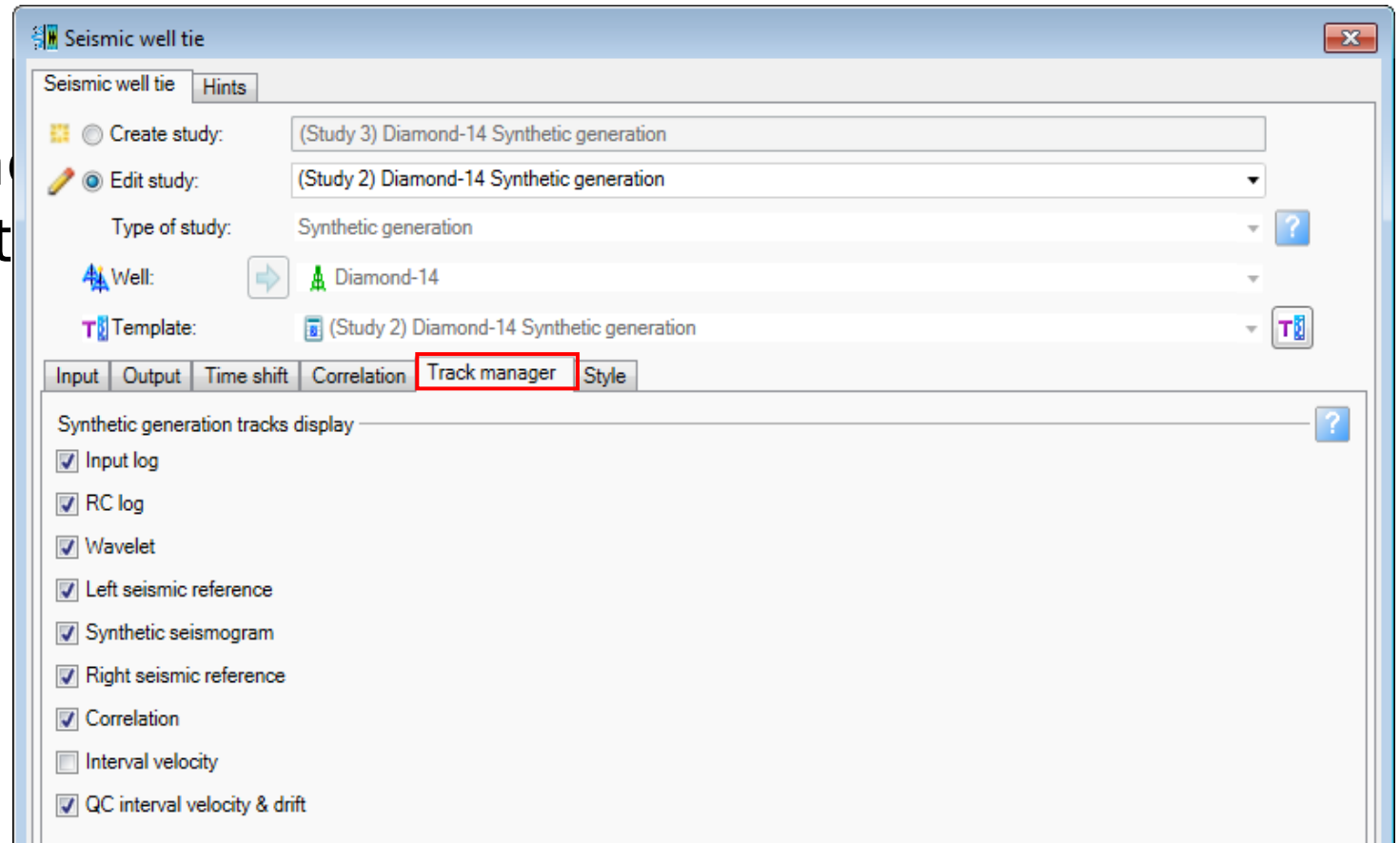
Run Reflection coefficient (RC) modeling (2)

3. Save your partial synthetic. On the **Output** tab of the **Seismic well tie** dialog box, click *Save* or select the *Auto save* check box.



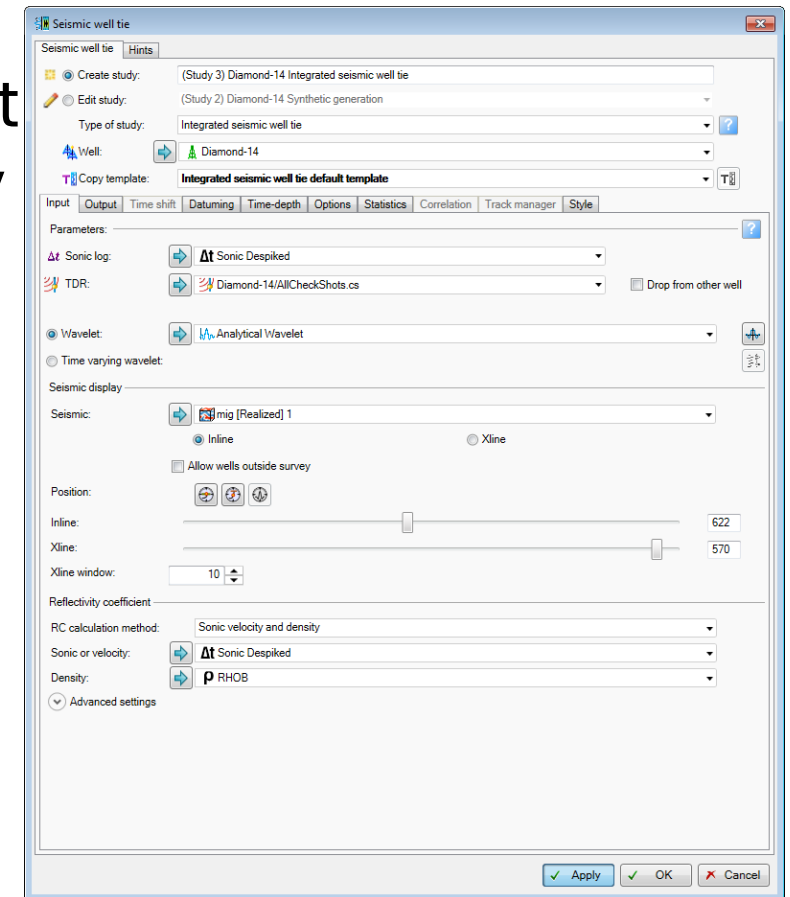
Track manager

- The **Track manager** tab provides a shortcut to help you manage the tracks that you want to show in the study.



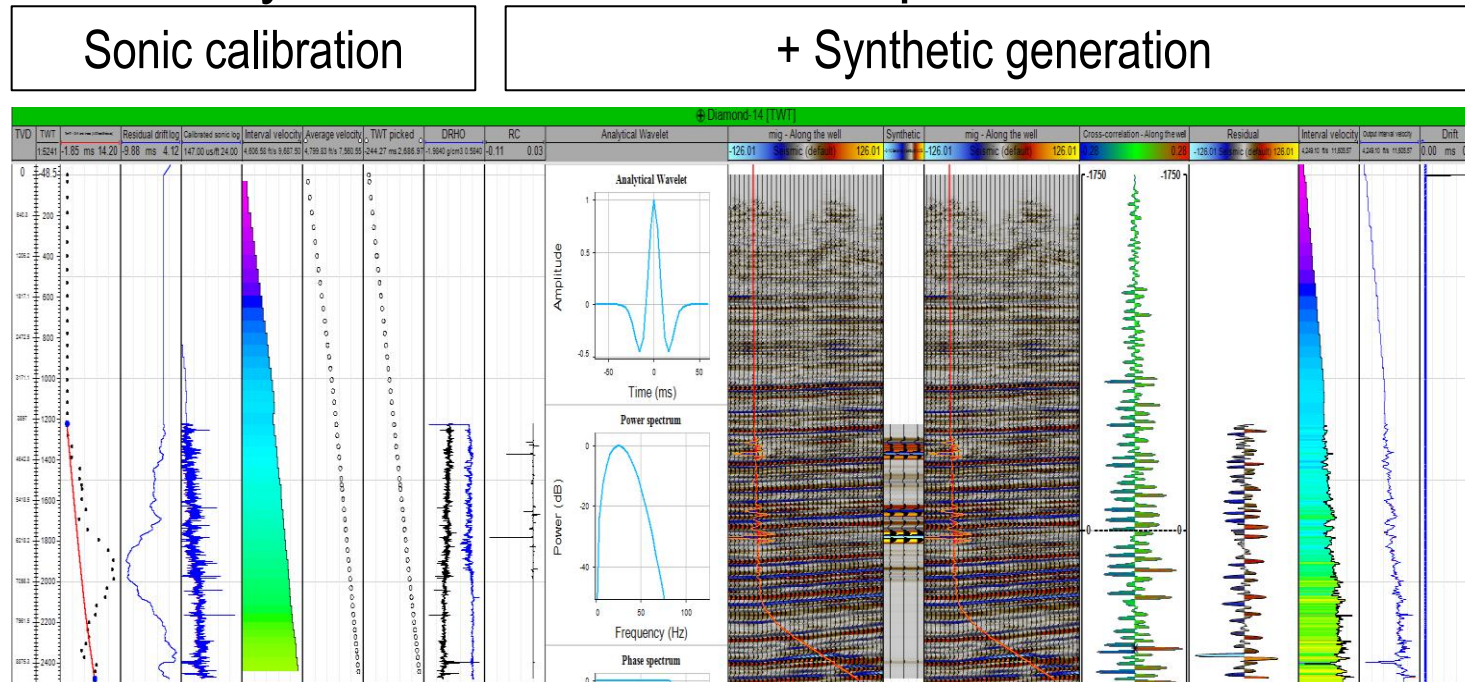
Integrated seismic well tie

- An integrated process in which Sonic Calibration Synthetic Generation is done simultaneously same **Well section window**.



Integrated seismic well tie study template

- Similar to the Sonic calibration and Synthetic generation, you can select a template for the study or use the default template.



Multi-well synthetic seismogram

- You have the option to create a synthetic generation study for multiple wells in a single **Well section window**.

Multi-well synthetic generation

Create multi-well synthetic study

Name: Multi-well synthetic

☐ Well section window:

☒ Wells:

Agate-H6

Diamond-14

Wavelet: Analytical Wavelet

Seismic: mig

Reflectivity coefficient

RC calculation method: Sonic velocity and density

Sonic or velocity: Calibrated sonic log

Density: DRHO

☐ Use Gardner's equation

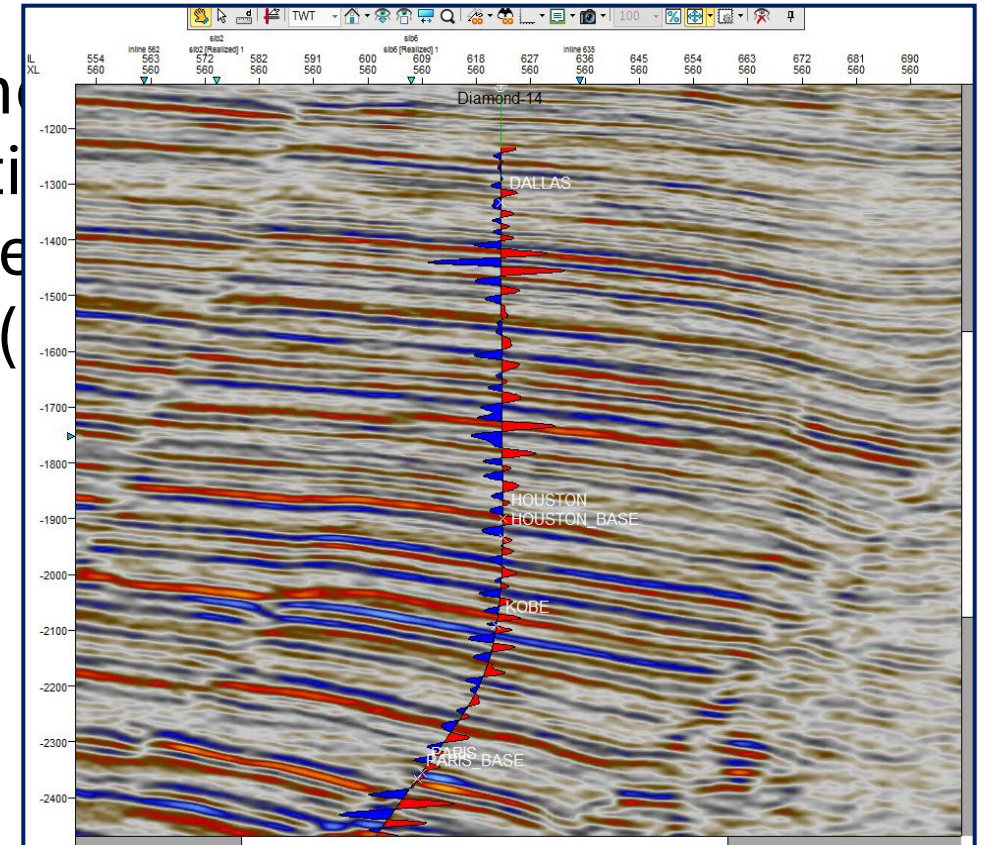
Constant: 0.23 imperial

Exponent: 0.25

OK Cancel

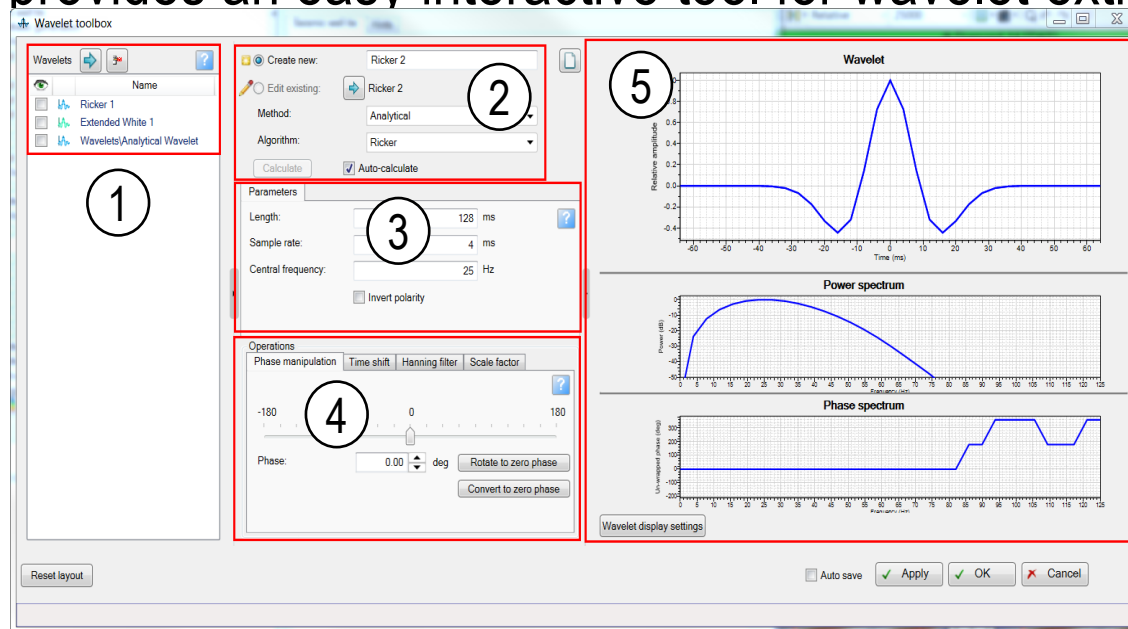
Synthetic seismogram in the Interpretation window

- In an **Interpretation window**, you can show the wellbore trajectory with the synthetic seismogram to identify the seismic reflections that represent the geological well tops (before starting horizon interpretation).



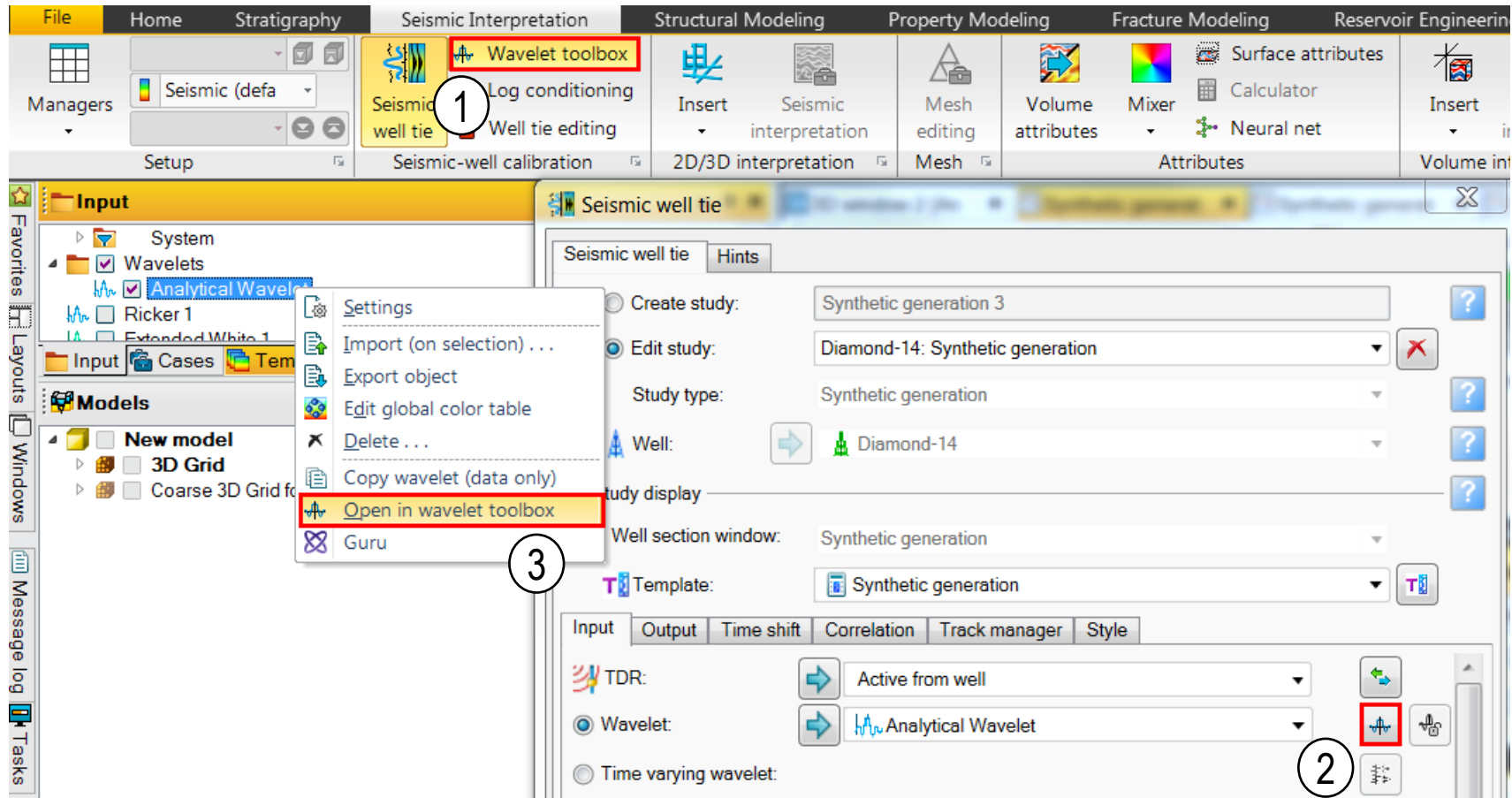
Wavelet toolbox

- The **Wavelet toolbox** integrates all related processes (Wavelet extraction, Wavelet Builder, and Wavelet viewer) in a single canvas.
- It provides an easy interactive tool for wavelet extraction.



- 1 Displayed Wavelet list
- 2 Method and algorithm
- 3 Parameters of extraction/generation
- 4 Operations
- 5 Visualizations

Access the Wavelet toolbox

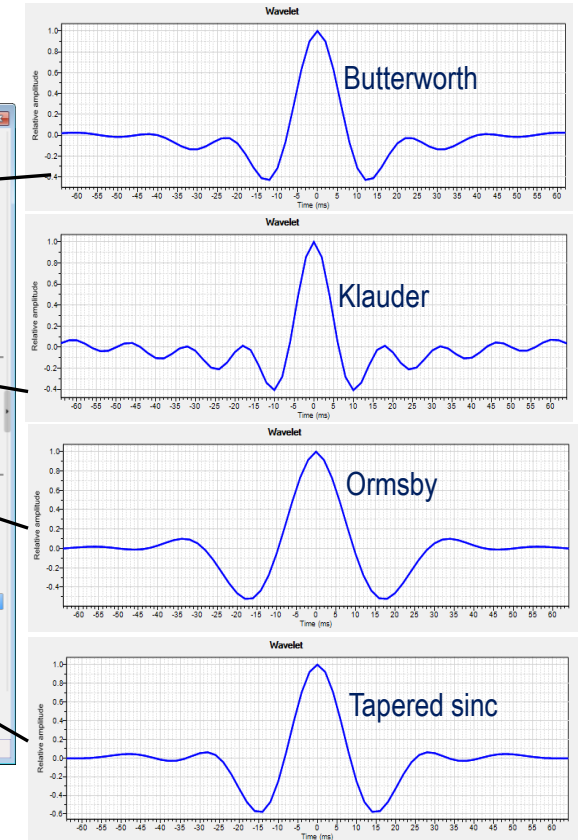
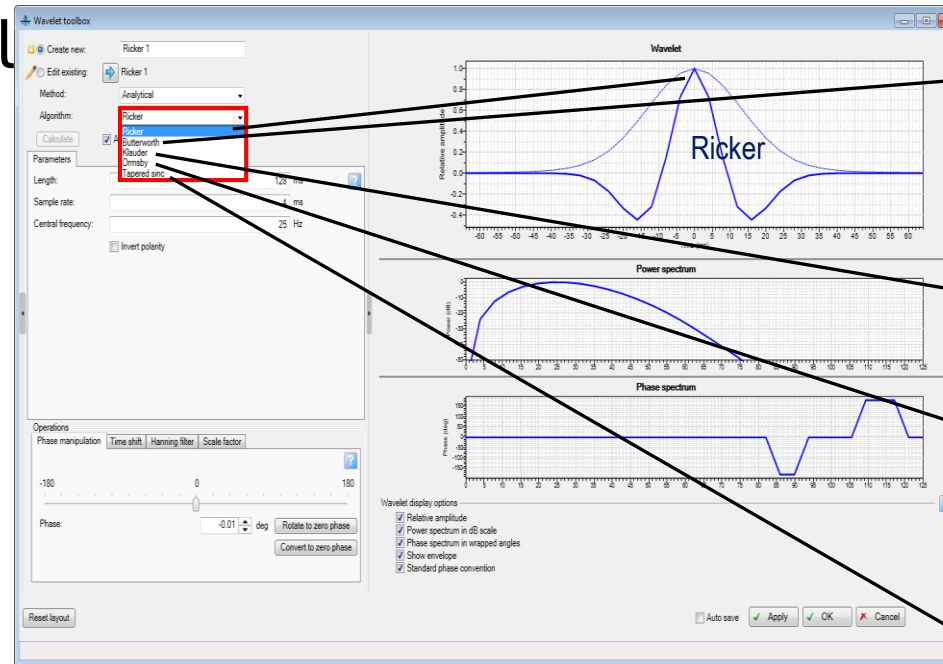


Types of wavelet extraction methods

- Analytical:
 - Ricker
 - Butterworth
 - Klauder
 - Ormsby
 - Tapered Sinc
- Statistical:
 - Extraction
 - Extended extraction
- Deterministic:
 - Extended White
 - Signal/Noise
 - Isis Frequency
 - Isis Time
- Multi wavelet: Wavelet Average
- Multi well

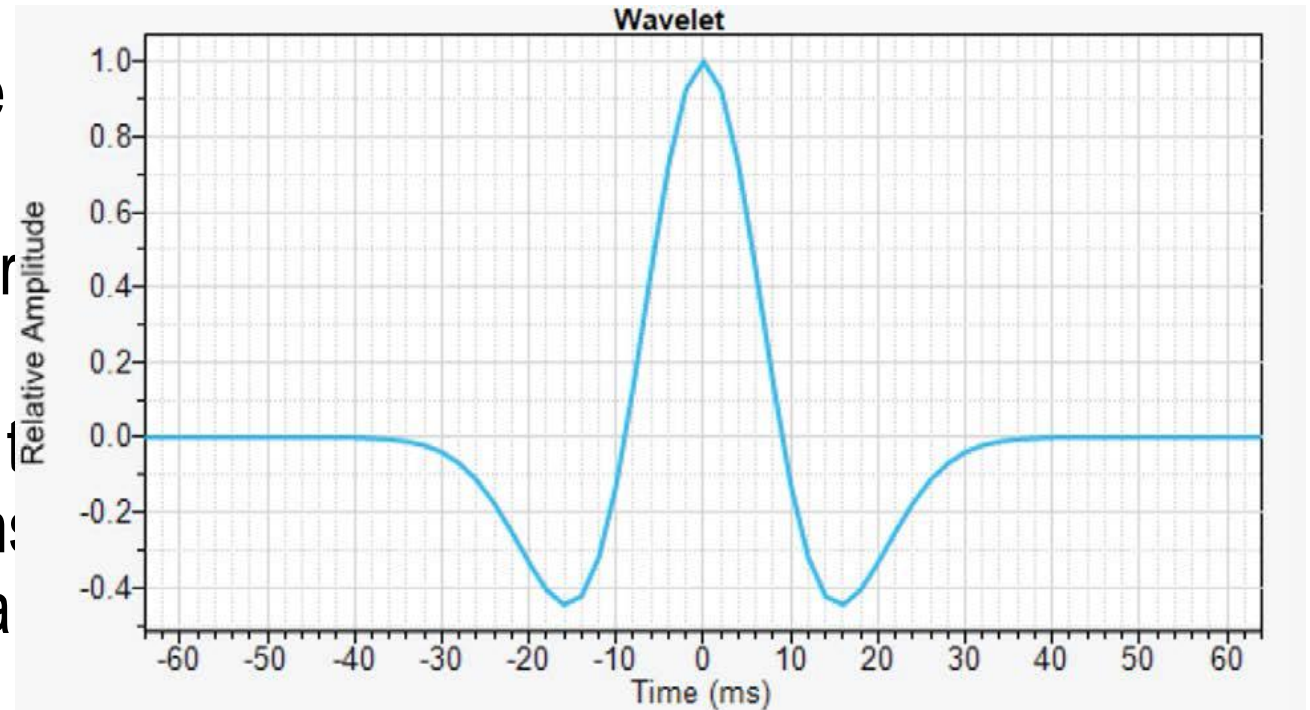
Analytical method

- Analytical wavelets are standard model wavelets.
- Five types are available



Ricker

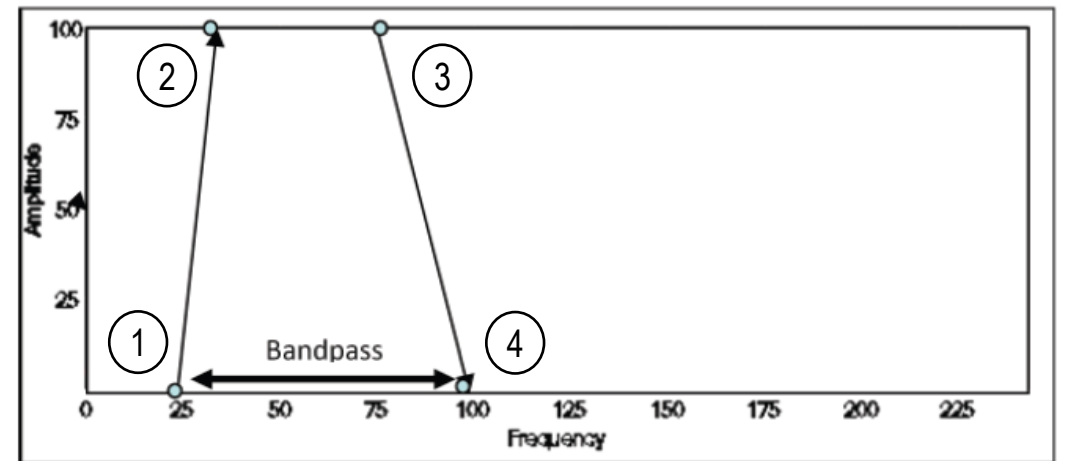
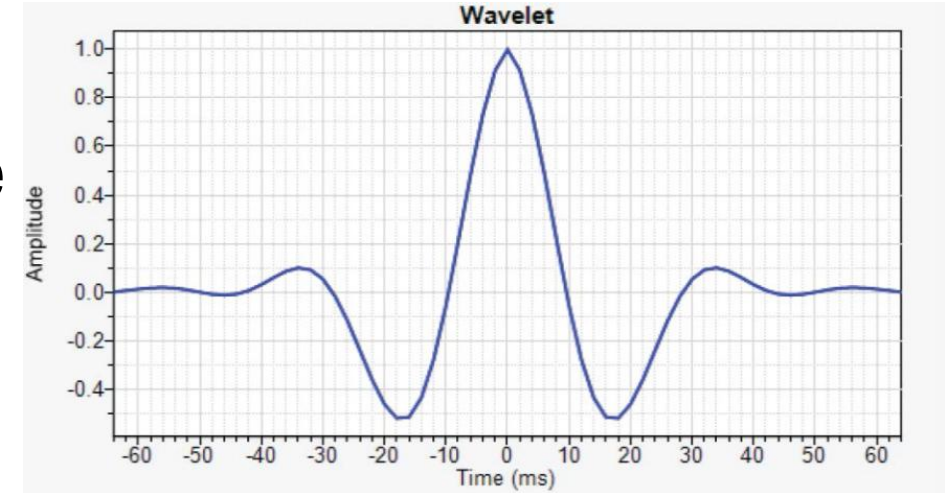
- A Ricker filter requires only the peak frequency.
- This filter commonly is used for synthetic modeling.
- No bandpass filter is involved; the frequency and phase spectrum are purely a function of the peak frequency input.



Ormsby

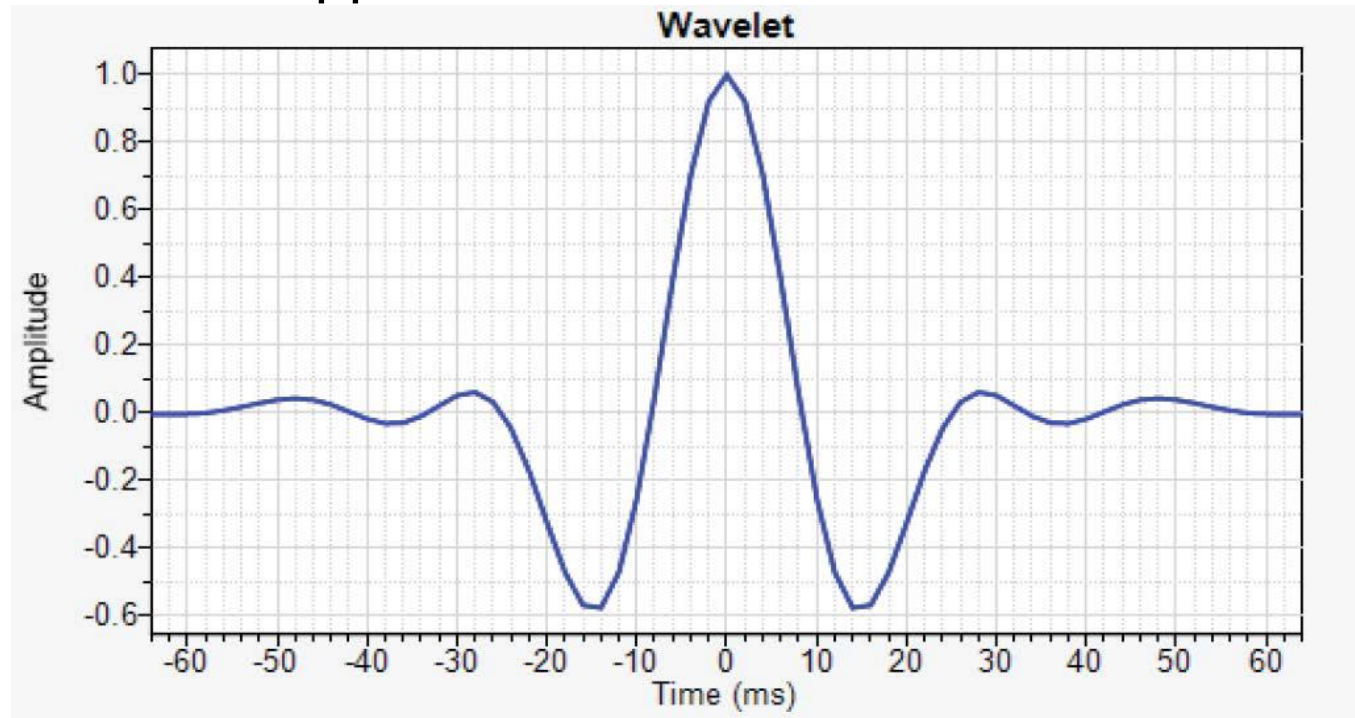
- The bandpass of an Ormsby filter can be described by using as many as four corner frequencies.

- 1 Low cut frequency
- 2 Low pass frequency
- 3 High pass frequency
- 4 High cut frequency



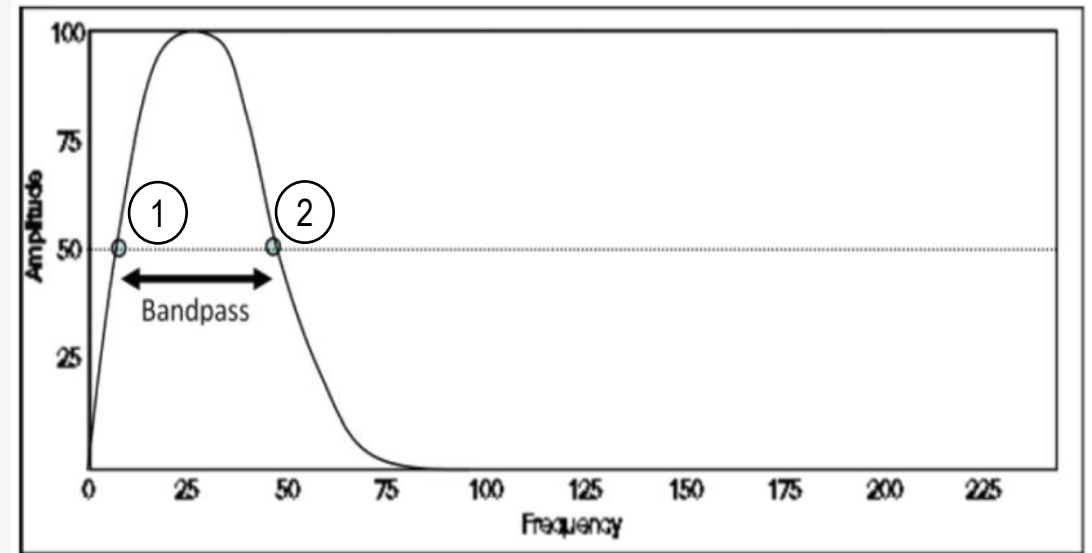
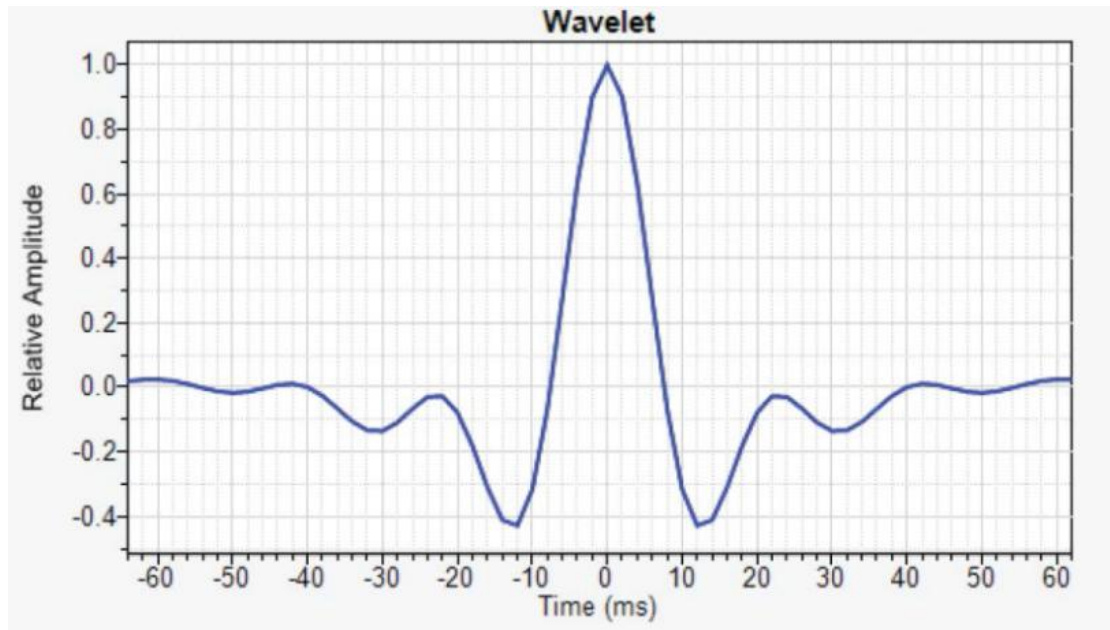
Tapered sinc

- A tapered sinc filter is defined by a low and a high cutoff similar to a Butterworth filter, but it applies no further filters. A Butterworth filter applies two slopes.



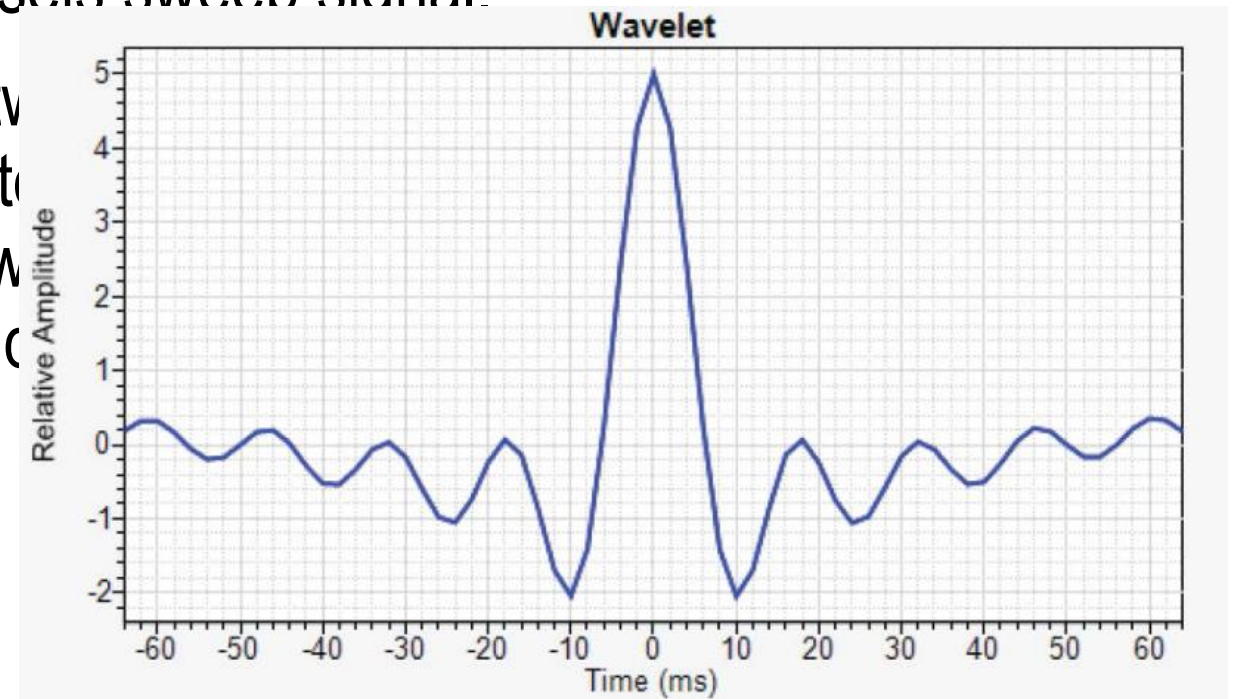
Butterworth

- The Butterworth bandpass consists of two cutoff frequencies taken at 3dB down from maximum power, or approximately half power (~50% on the amplitude scale). In the figure, frequencies are at 10Hz and 50Hz. The Butterworth filter also requires two slopes.



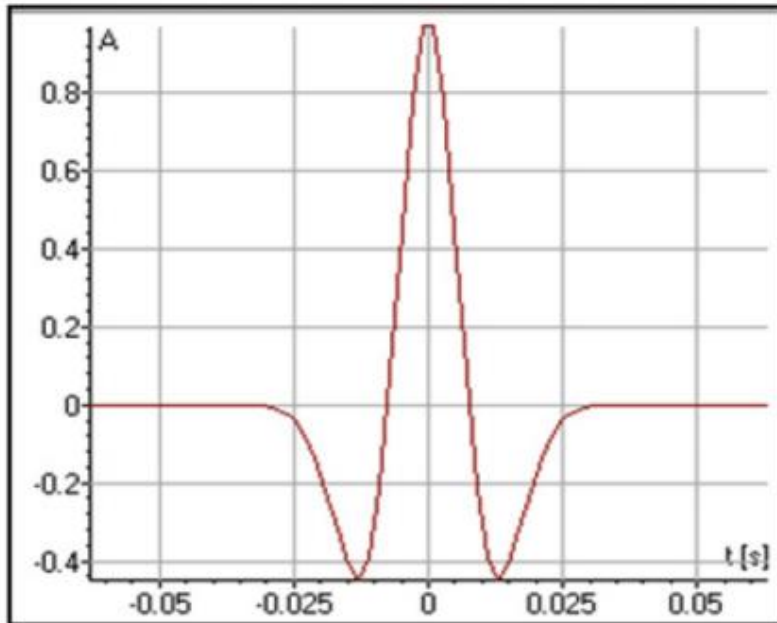
Klauder

- An analytical approximation of the Klauder wavelet is computed through autocorrelation of an actual vibroseis sweep signal.
- A Klauder wavelet is defined by two frequency cutoff values: a low cutoff and a high cutoff. The figure shows these frequencies set at 10Hz and 70Hz.

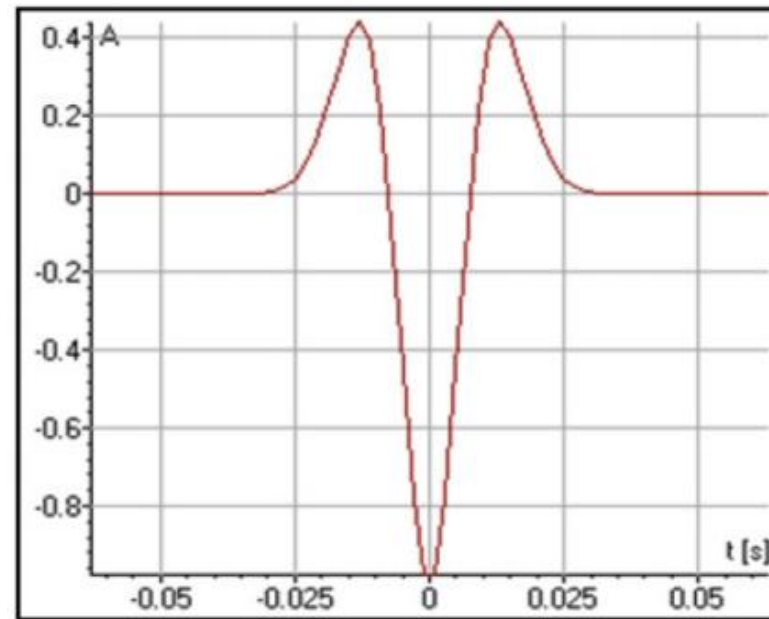


Wavelet phase

- Convention for a zero-phase wavelet for the USA and Europe are opposite in phase



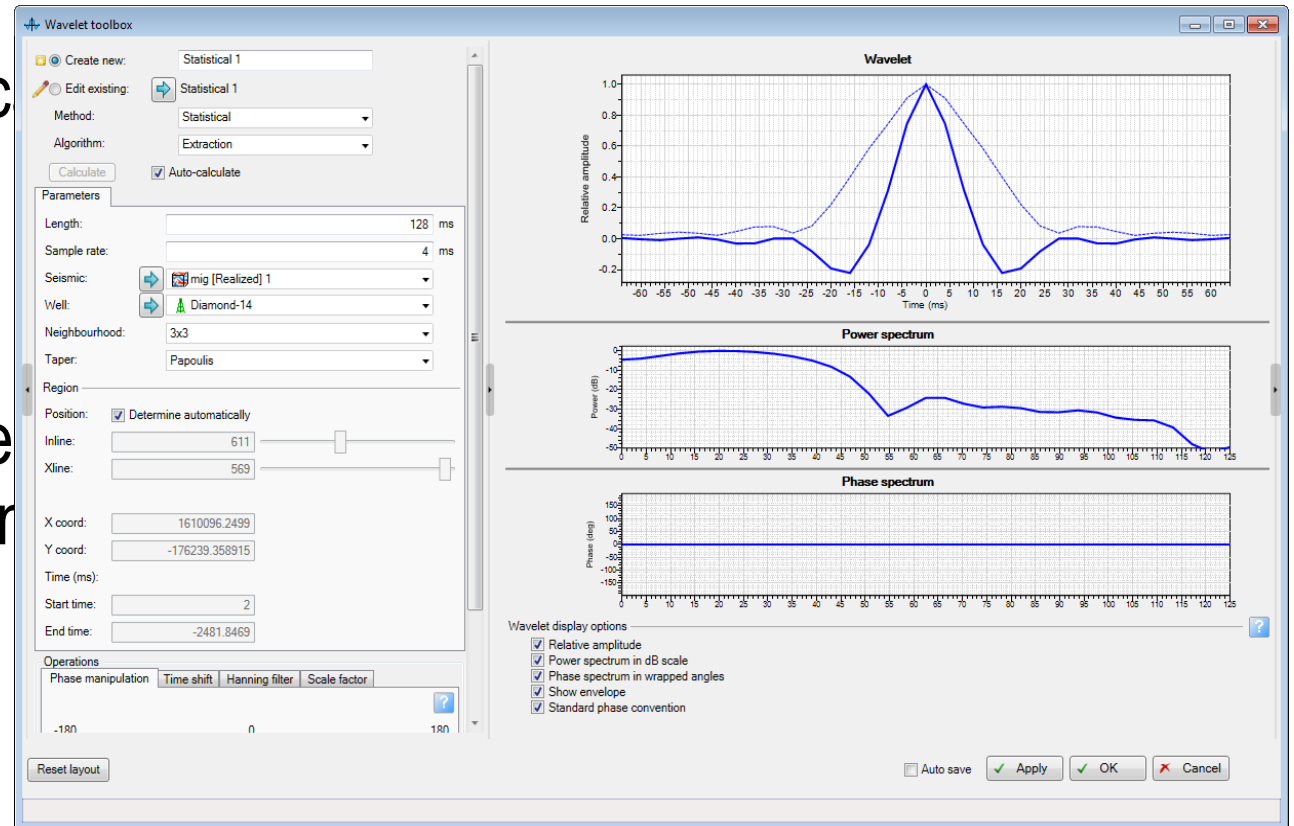
USA zero-phase wavelet



European zero-phase wavelet

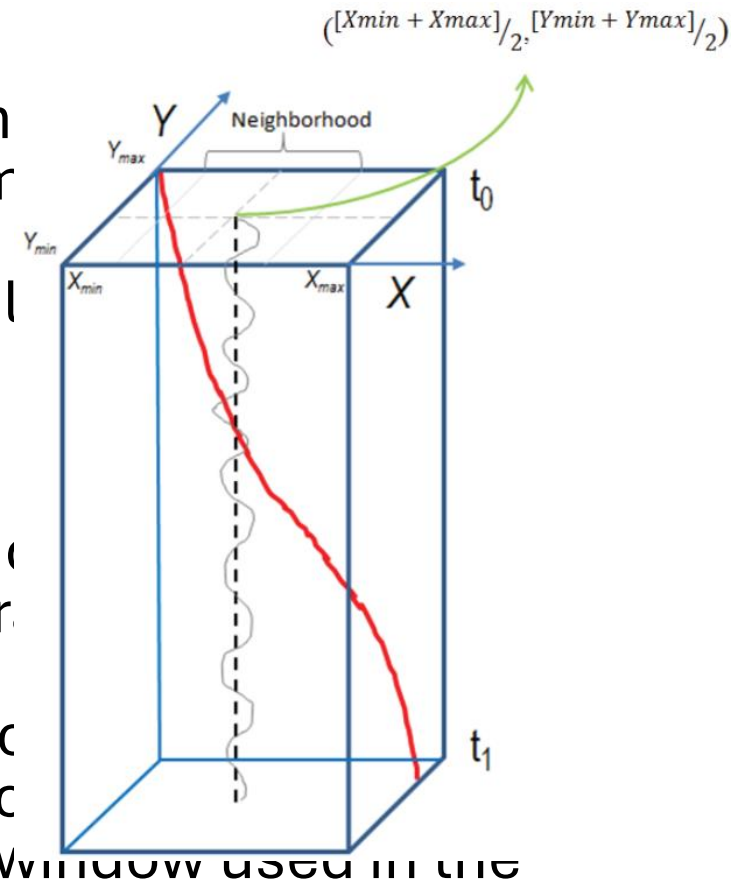
Statistical method

- It is possible to access statistic extraction even if no sonic log exists for the borehole.
- This method assumes that the embedded wavelet is the same as the truncated autocorrelation of the seismic trace.



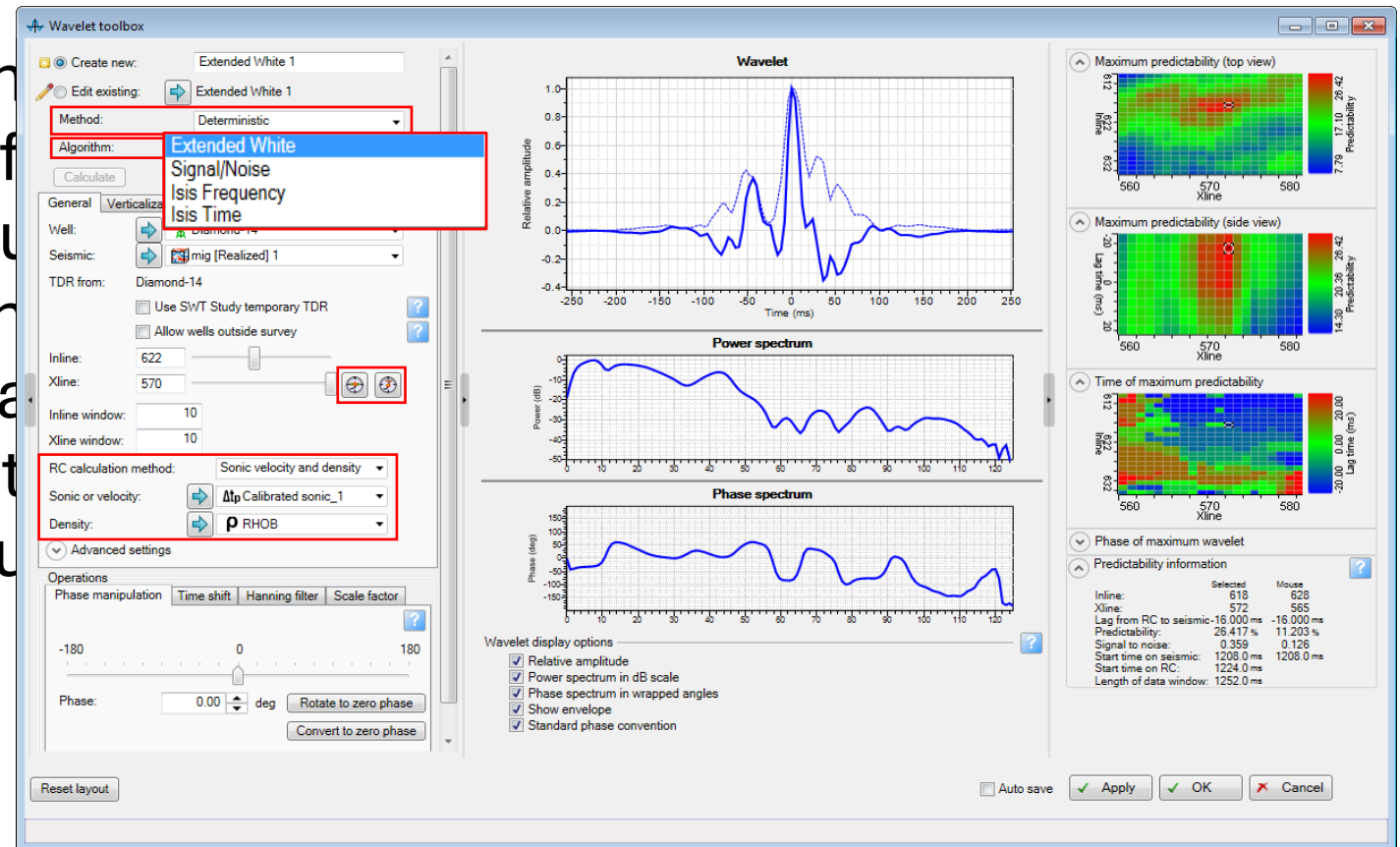
Principles of the Statistical method

- Given a borehole trajectory, this method extracts both minimum and maximum values of X and Y: (Xmin, Ymin) and (Xmax, Ymax).
- The auto position (determined automatically) is calculated as the mean between them:
 - $([Xmin+Xmax]/2, [Ymin+Ymax]/2)$
- It defines the position of the central trace for the extraction.
- Neighborhood (1x1, 3x3, 5x5) defines the number of traces used for the average.
- Unlike the Extraction algorithm, the Extended extraction has no limit on the number of traces used in the extraction window. It provides different options to limit the time window used in the extraction.



Deterministic method

- For this method, a seismic volume and input logs of interest are required. You can change the position of the extraction location interactively based on predictability, to optimize the wavelet to u

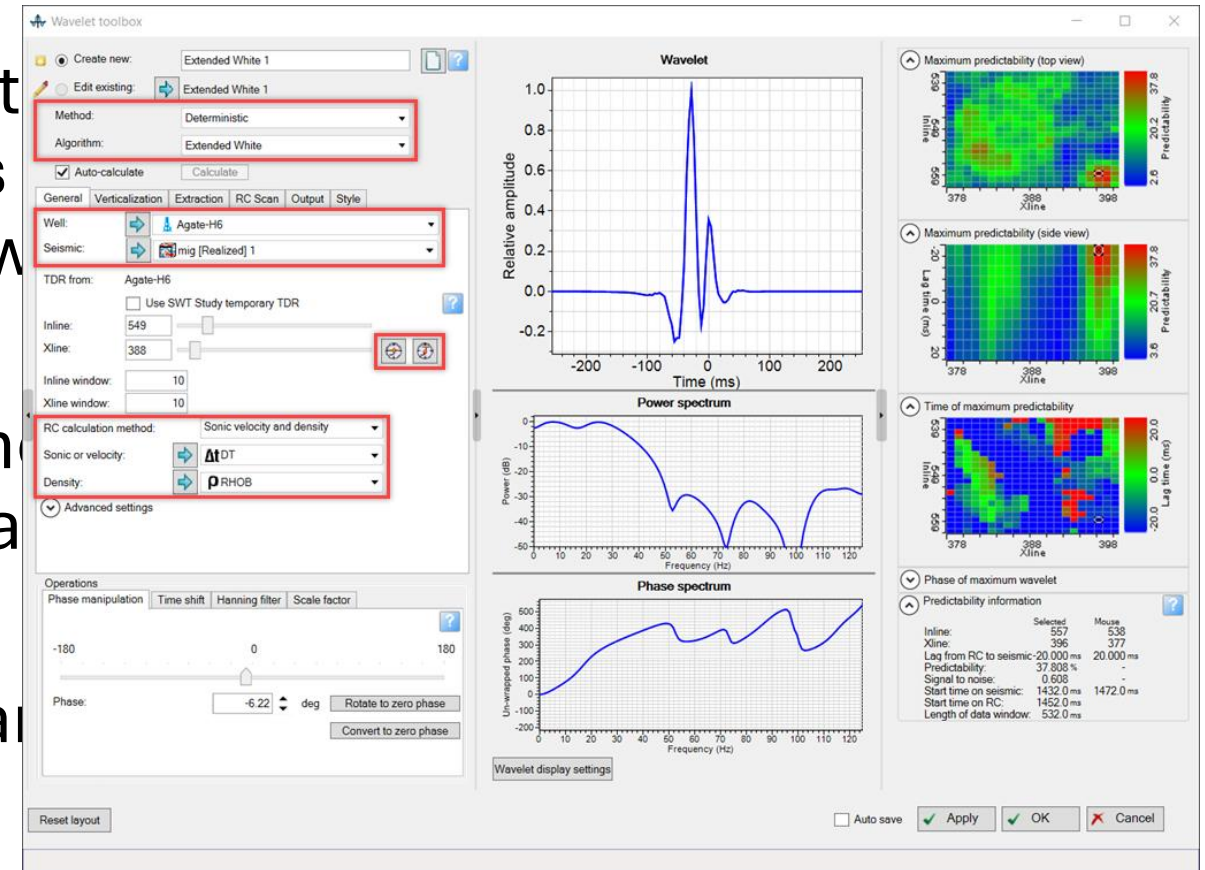


Extended White

- The Extended Roy White wavelet extraction method decomposes tie and wavelet extraction into two components:

■ Determination of optimal time between well log reflectivity and seismic amplitude

■ Determination of the dominant

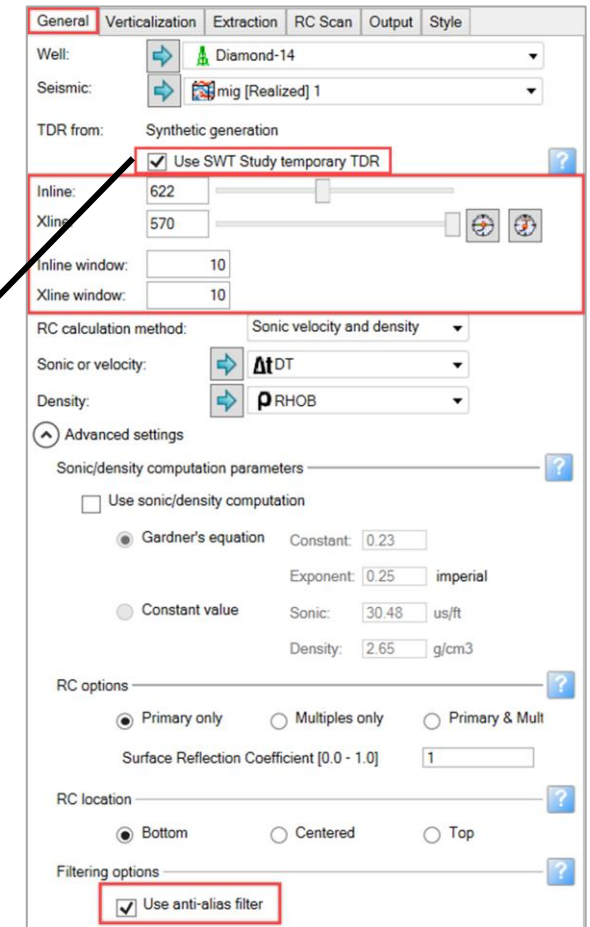
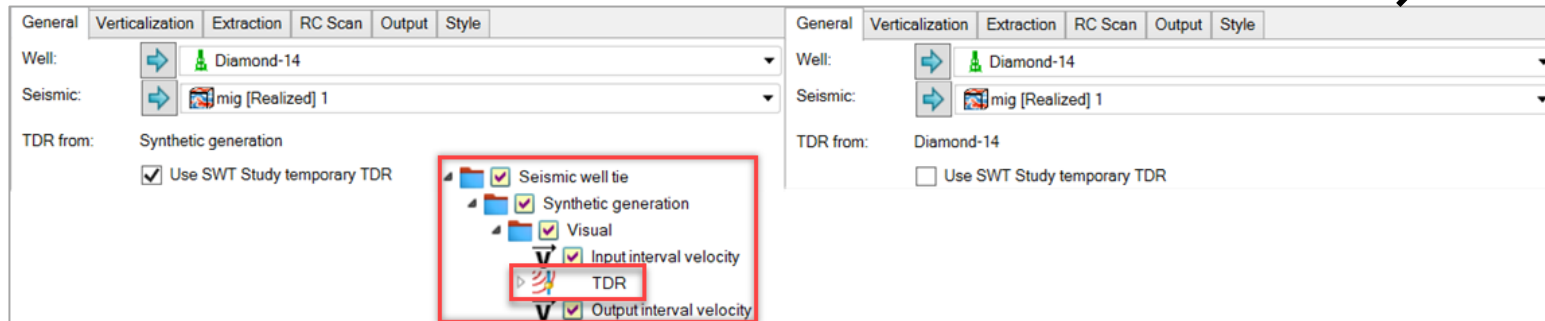


Settings for Extended White

- After you select the deterministic method of wavelet extraction in the **Wavelet** toolbox, six tabs allow you to better define wavelet extraction.
 - General
 - Verticalization
 - Extraction
 - RC Scan
 - Output
 - Style

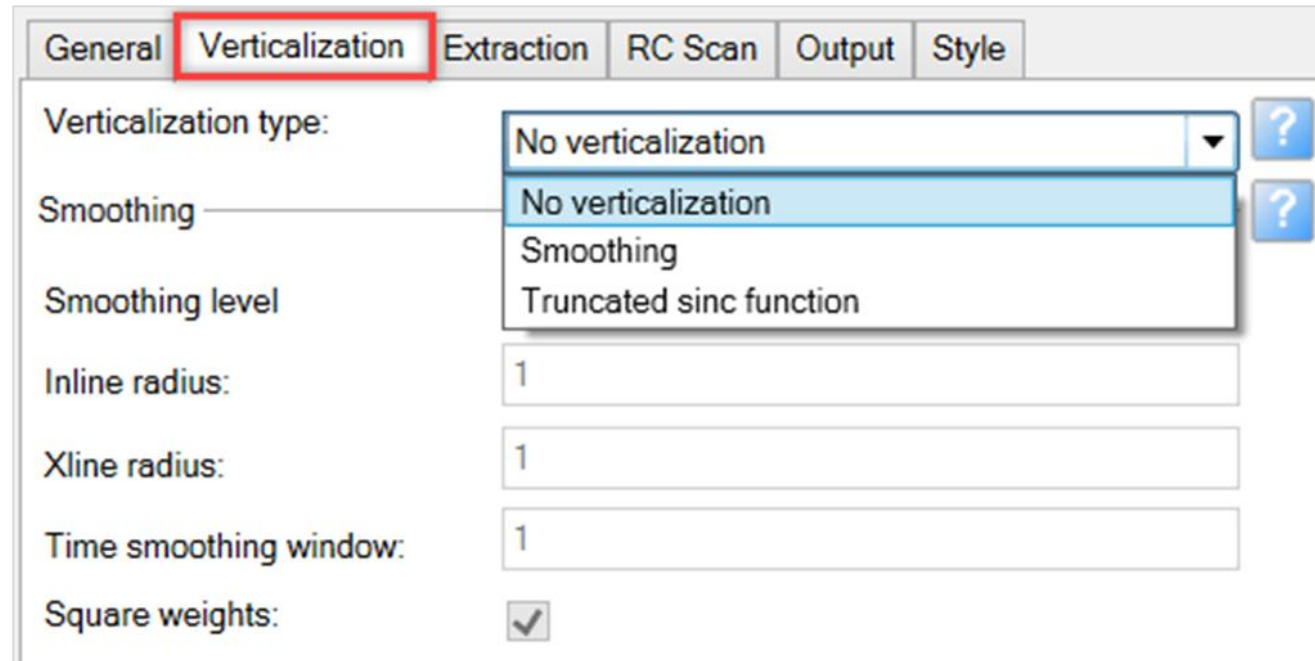
General tab

- In an active **Well section window**, you can use a temporary Time Depth Relationship (TDR) or the well TDR to extract a deterministic wavelet.
- If you use the SWT temporary TDR, any change from the bulk shift or stretch and squeeze process in the study is used by the **Wavelet** toolbox for the extraction.



Verticalization tab

- A deviated trajectory often crosses several different seismic traces; the **Verticalization** tab offers options for merging them to extract a wavelet.



The screenshot shows a software interface with a tabbed menu at the top. The tabs are 'General', 'Verticalization', 'Extraction', 'RC Scan', 'Output', and 'Style'. The 'Verticalization' tab is selected and highlighted with a red rectangle. Below the tabs, the 'Verticalization type:' dropdown menu is open, showing three options: 'No verticalization' (selected), 'Smoothing', and 'Truncated sinc function'. To the right of the dropdown are two blue question mark icons. Below the dropdown, there are input fields for 'Smoothing level', 'Inline radius:', 'Xline radius:', and 'Time smoothing window:', all containing the value '1'. At the bottom, there is a checkbox for 'Square weights:' which is checked.

Tab	Verticalization type	Smoothing level	Inline radius	Xline radius	Time smoothing window	Square weights
General						
Verticalization	No verticalization		1	1	1	☒
Extraction						
RC Scan						
Output						
Style						

Extraction tab

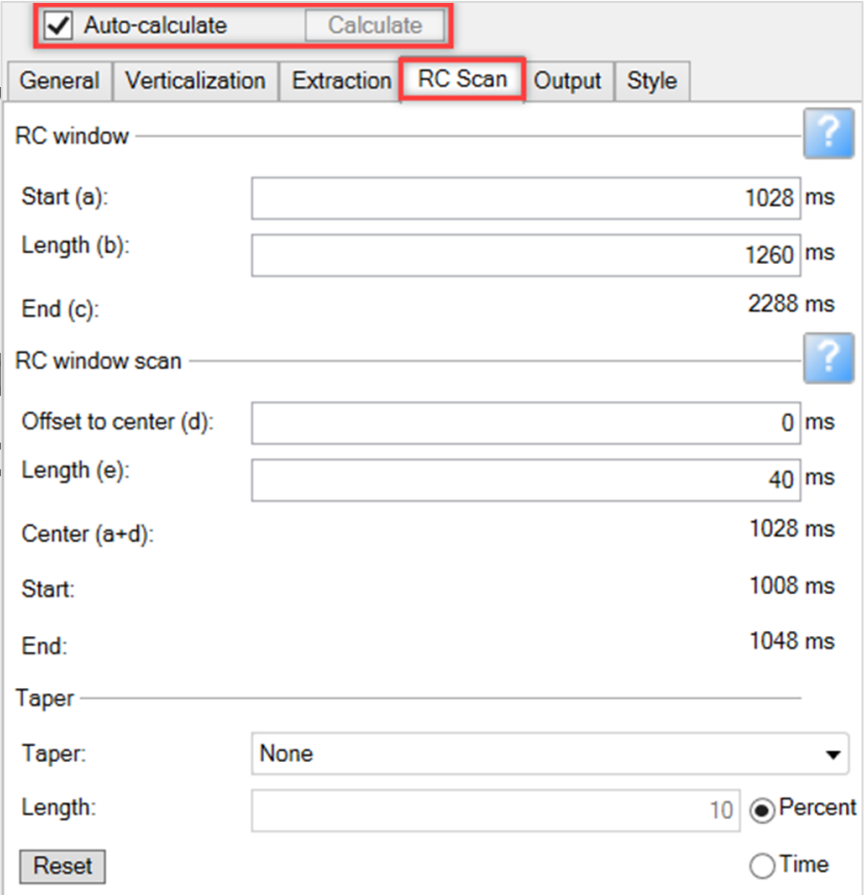
- Select the *User defined* check boxes to customize the length of the taper applied to the cross-correlation that is computed and used for wavelet estimation.

The screenshot shows a software window with several tabs: General, Verticalization, Extraction (highlighted with a red box), RC Scan, Output, and Style. The 'Extraction' tab is active, displaying the following parameters:

- Parameters** (with a help icon ?):
 - Length of extraction xcor: 256 ms ☐ User defined
 - Length of extraction xcor taper: 128 ms ☐ User defined
 - ☐ Rescale amplitudes
- White Noise** (with a help icon ?):
 - Percentage of white noise: 1.00 %
- Reset** button

RC Scan tab

- Contains the options for setting the time window for the wavelet extraction process.
- Activate the *Auto-calculate* option to update the wavelet automatically as parameters are changed.
- Alternatively, use the *Calculate* option after inserting the desired parameters.



The screenshot shows the 'RC Scan' tab of a software interface. At the top, there is a checkbox labeled 'Auto-calculate' which is checked, and a 'Calculate' button. Below this are tabs for 'General', 'Verticalization', 'Extraction', 'RC Scan' (which is selected), 'Output', and 'Style'. The 'RC Scan' section contains several input fields and labels:

- RC window**: A section with a help icon (?). It includes:
 - Start (a): 1028 ms
 - Length (b): 1260 ms
 - End (c): 2288 ms
- RC window scan**: A section with a help icon (?). It includes:
 - Offset to center (d): 0 ms
 - Length (e): 40 ms
 - Center (a+d): 1028 ms
 - Start: 1008 ms
 - End: 1048 ms
- Taper**: A section with a help icon (?). It includes:
 - Taper: A dropdown menu set to 'None'.
 - Length: 10, with radio buttons for 'Percent' (selected) and 'Time'.

At the bottom left of the 'RC Scan' section is a 'Reset' button.

Output tab

- Deterministic method can be used to generate Reflectivity, Acoustic Impedance, and the Dephase Operator as outputs. (Dephase Operator is a special purpose wavelet used to
- You also can save a wavelet snapshot.

General Verticalization Extraction RC Scan **Output** Style

Output parameters ?

RC Reflectivity: New reflectivity

AI Acoustic impedance: New acoustic impedance

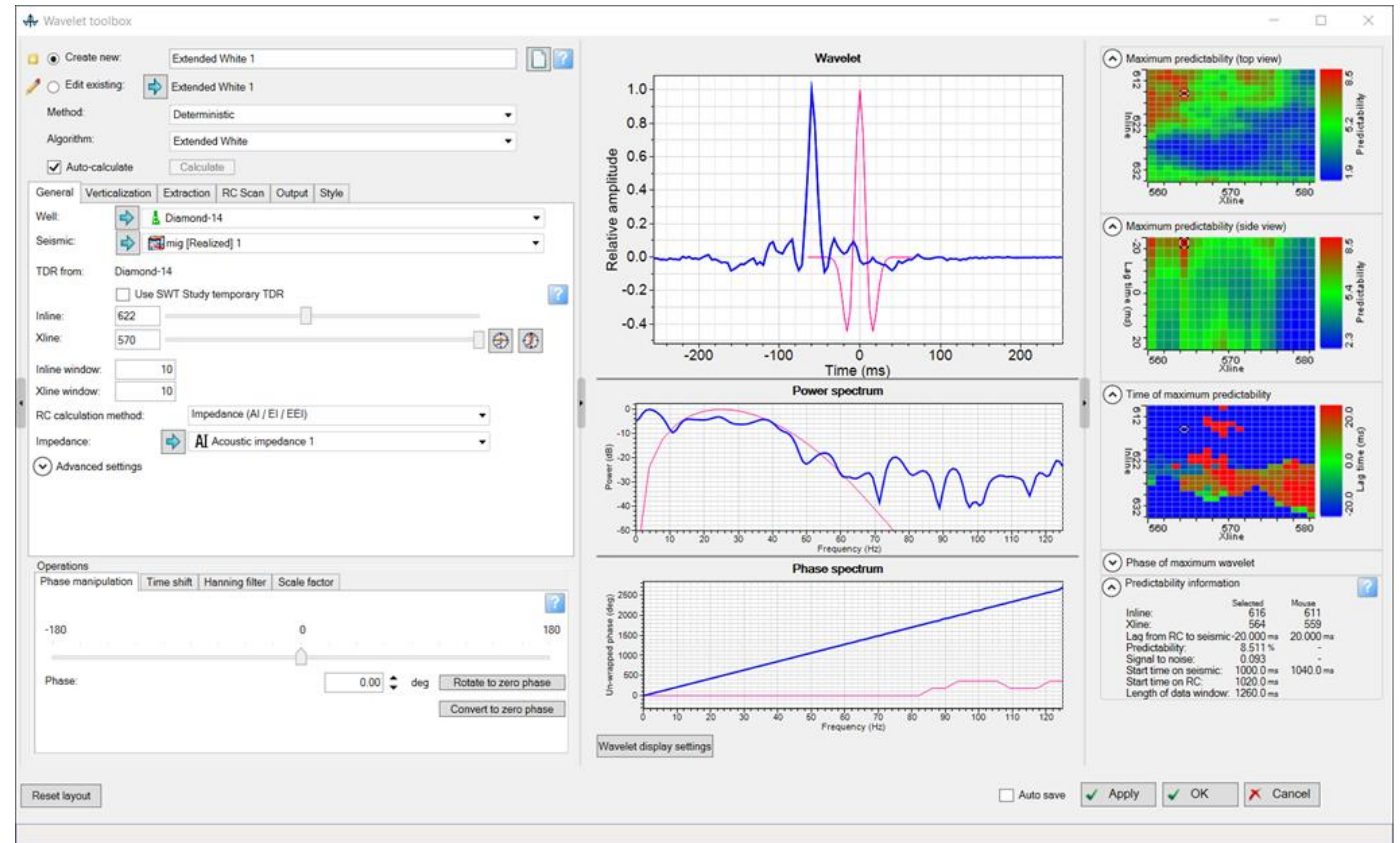
Dephase operator: New dephase operator

Output wavelet snapshot ?

Wavelet: Diamond-14-mig [Realized] 1-Extended White 1

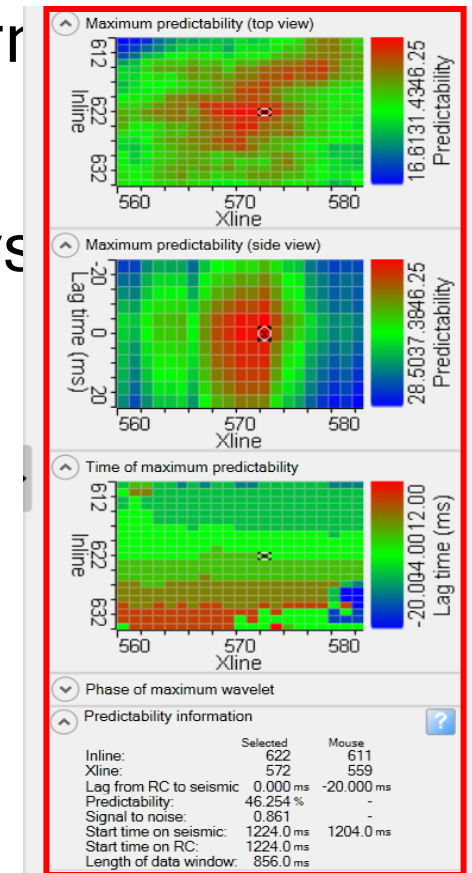
Wavelet extraction display (1)

- A **Predictability** section appears when you set all of the input parameters.
- Any change in the inputs or in the extraction parameters automatically updates the Predictability.



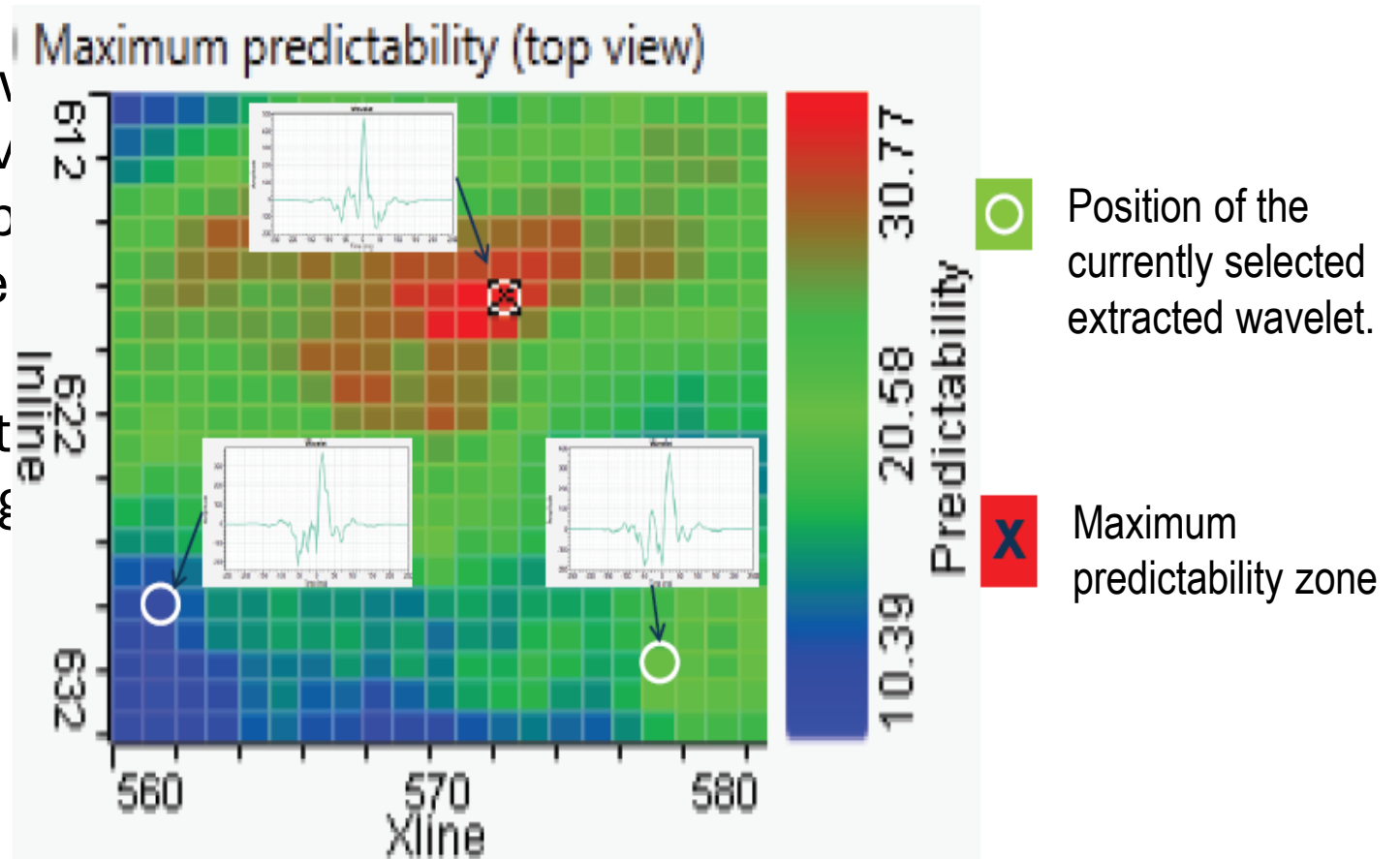
Wavelet extraction display (2)

- Predictability in a Seismic well tie is used to determine optimal wavelet.
- The windows show different Predictability displays
 - Maximum Predictability (top view)
 - Maximum Predictability (side view)
 - Time of maximum predictability
 - Phase of maximum wavelet
 - Predictability information



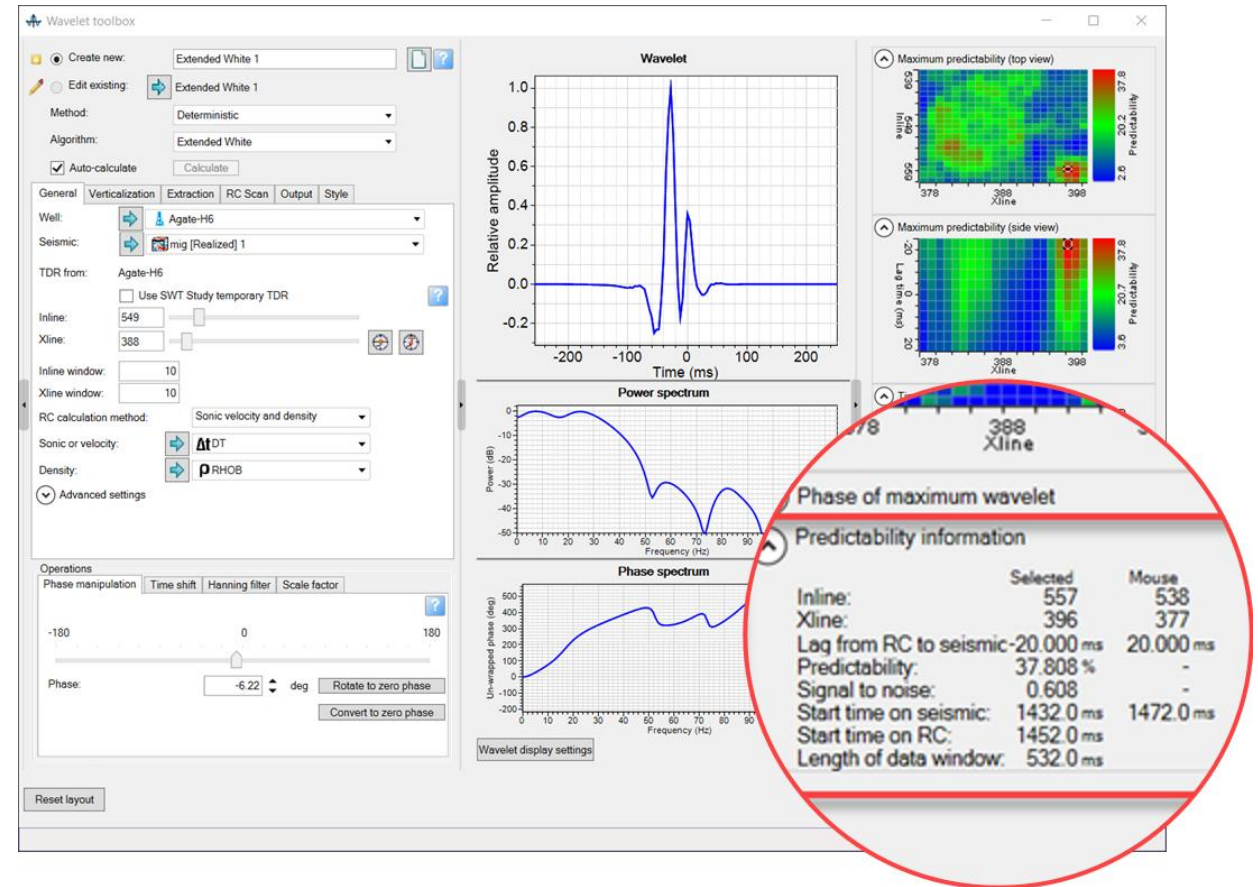
Wavelet extraction display (3)

- Initially, a white circle shows the currently selected wavelet at the maximum predictability of the cube of the extracted wavelet.
- The interpreter can move the extraction point by clicking on any predictability display.
- A black **X** shows the maximum predictability zone.



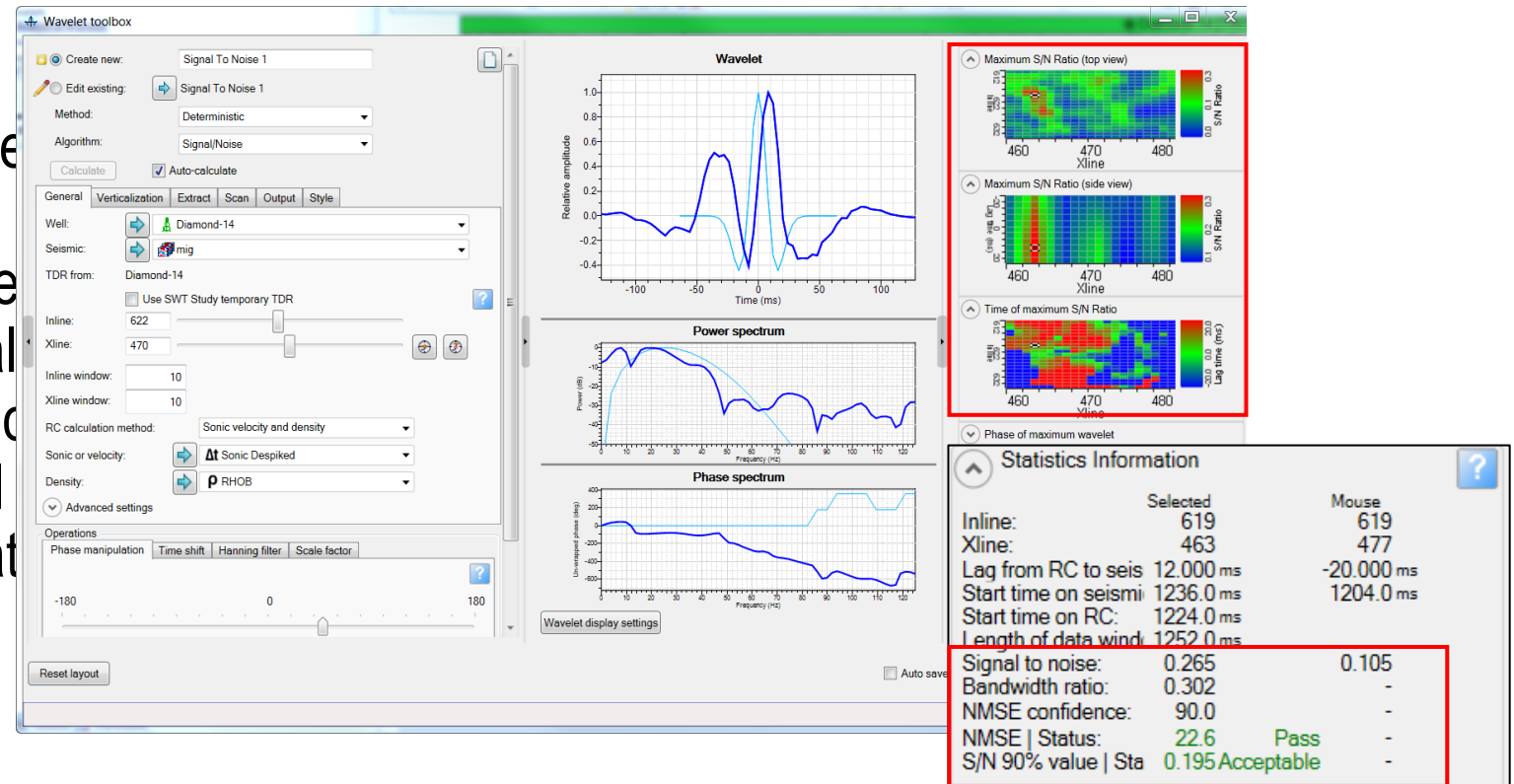
Predictability and correlation

- Predictability is a measure of the similarity of the underlying reflectivity.
- Simple cross-correlation methods do not benefit from these features.



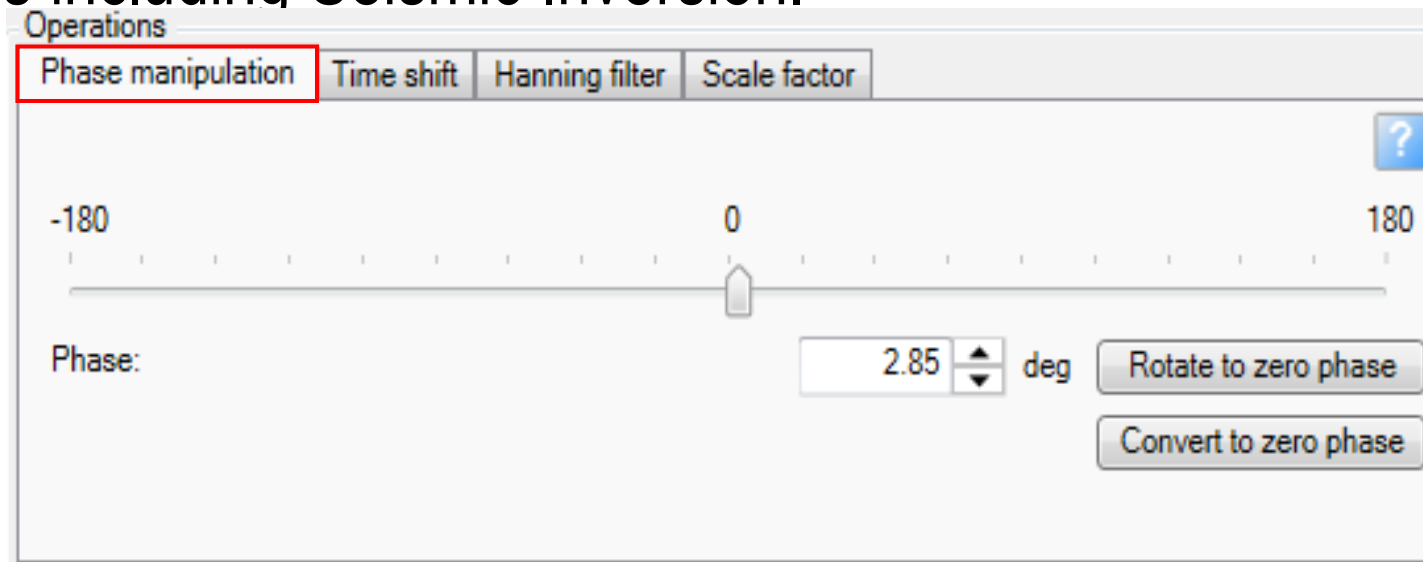
Signal/Noise methods for deterministic wavelets

- Based on Roy White, the Signal/Noise algorithm provides the best location based on Signal/Noise (S/N); that is, the ratio of the power in the signal (power in the smooth synthetic) over the power in the residual the match with the seismic data



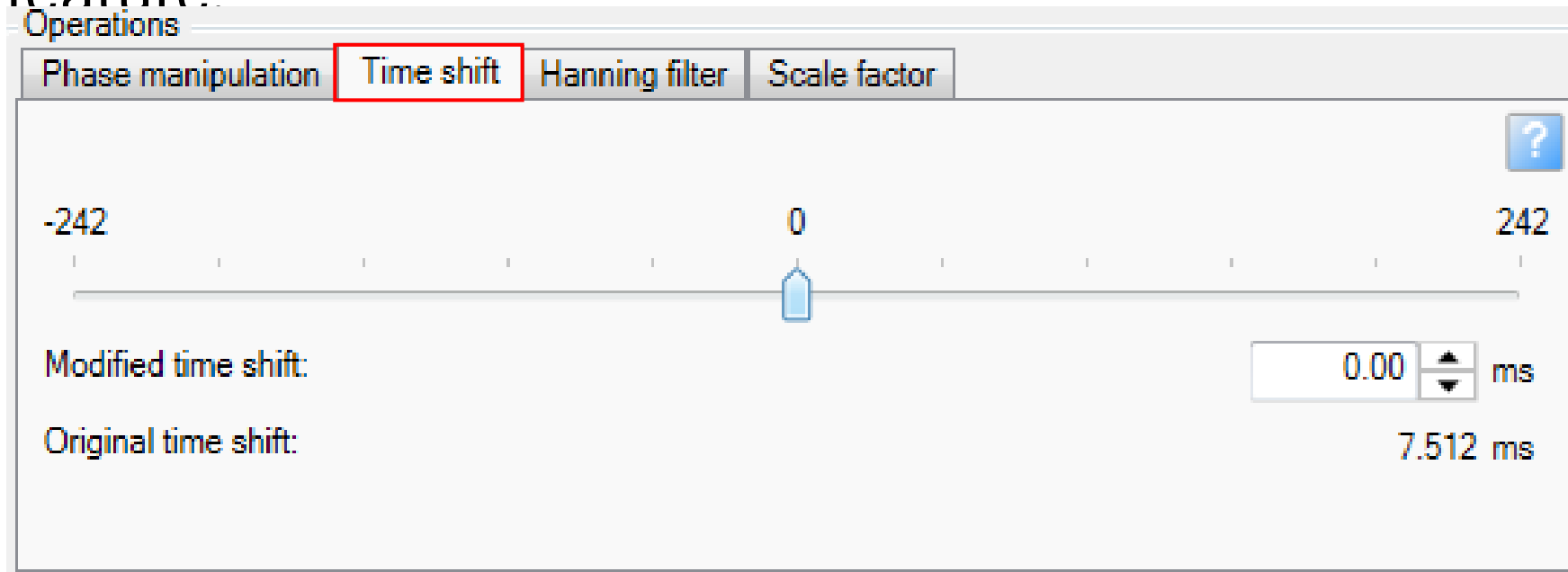
Phase manipulation

- Understanding the phase of the seismic data is important.
- Zero phase seismic data is essential for many geophysical analysis techniques including Seismic Inversion.



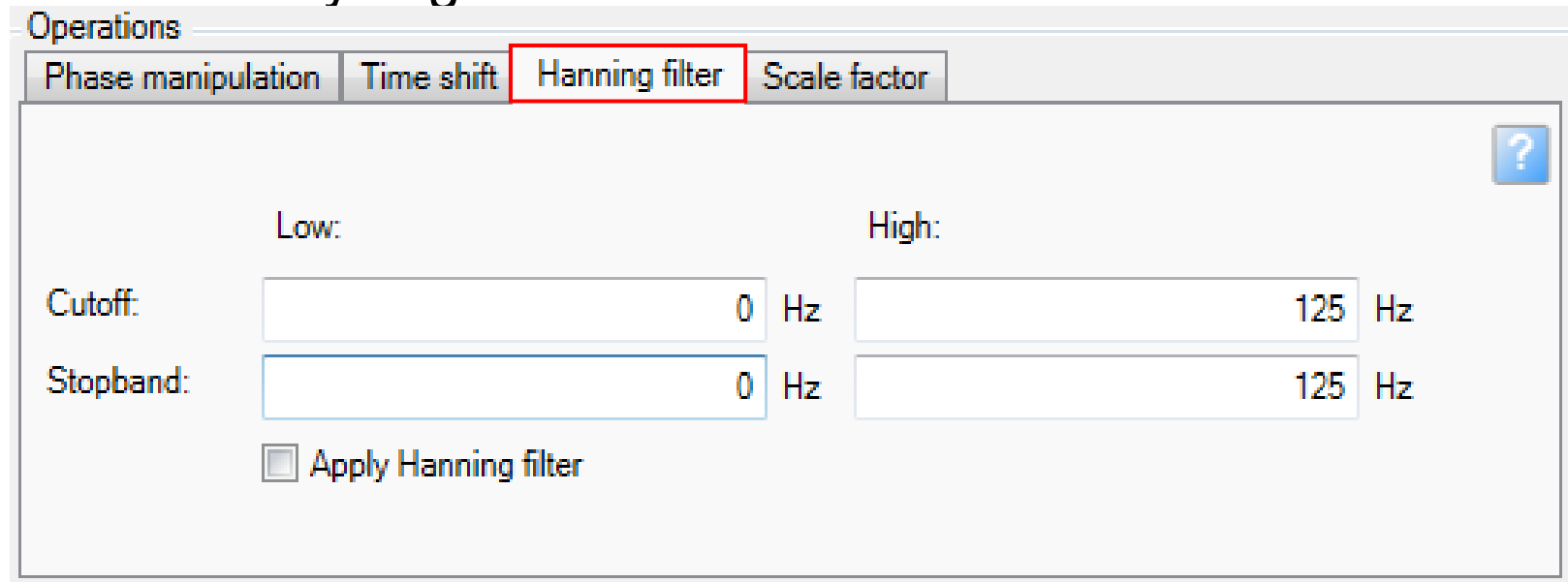
Time shift

- A bulk shift can be applied to the entire wavelet using the Time shift feature.



Hanning filter: Seismic data frequency characteristics

- It helps to evaluate if a wavelet extracted for a deeper interval is applicable for a shallower interval. This evaluation is done by filtering out the lower frequencies and analyzing tie at a shallow interval.



Operations

Phase manipulation Time shift **Hanning filter** Scale factor

Low: High:

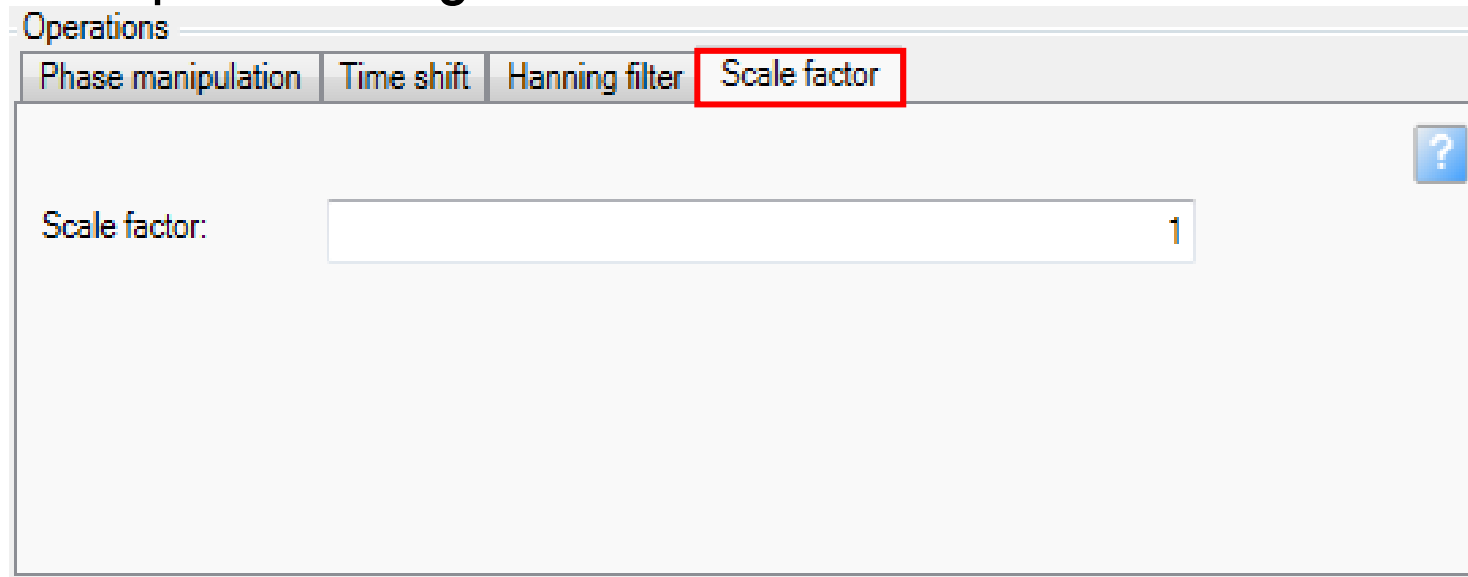
Cutoff: 0 Hz 125 Hz

Stopband: 0 Hz 125 Hz

☐ Apply Hanning filter

Scale factor

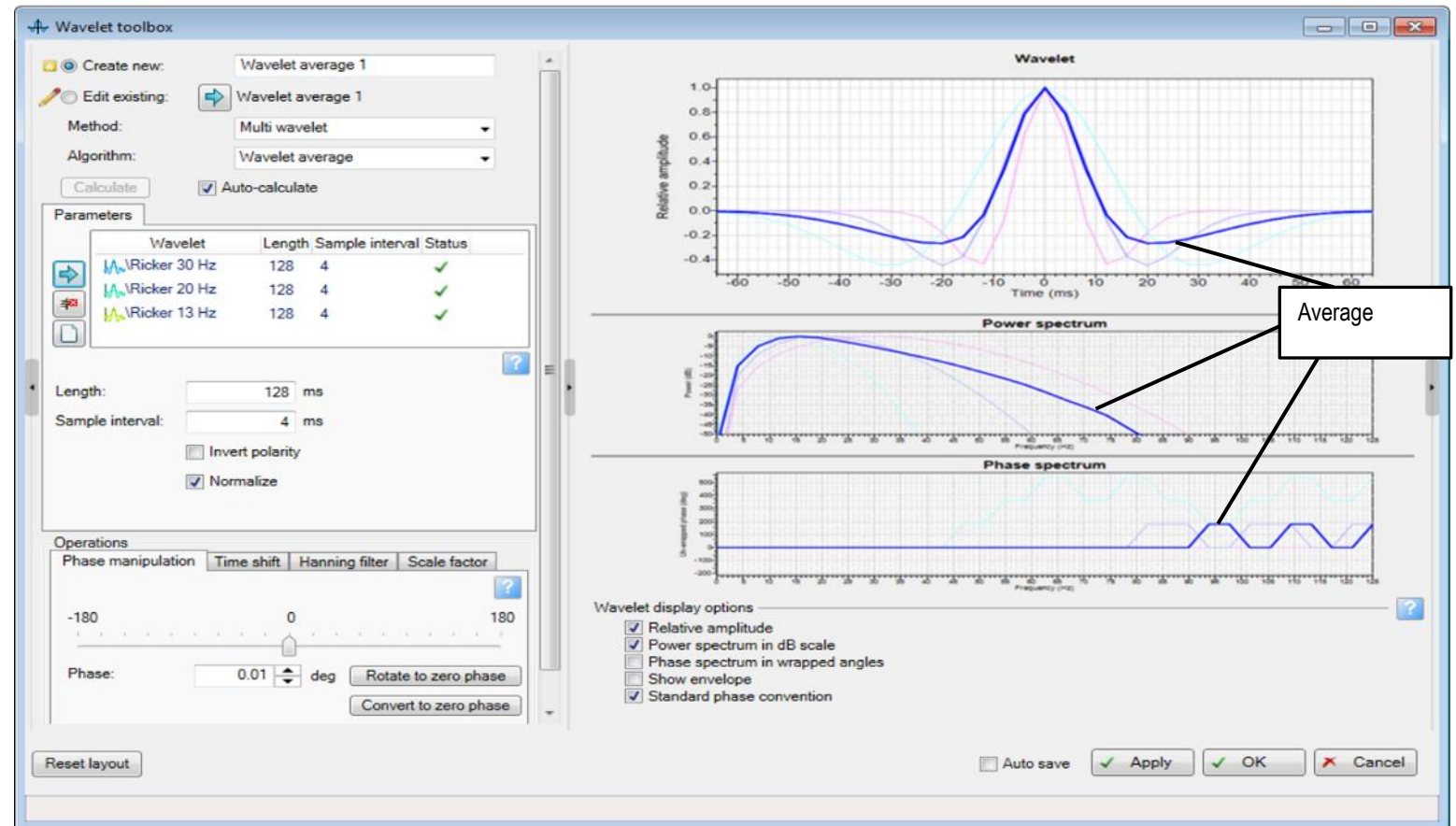
- Scales the amplitude of the wavelet.
- For the analytical and statistical normalized method of wavelet extraction in **Petrel**, the amplitude range varies from -1 to 1.



The screenshot shows a software interface window titled "Operations". It contains four tabs: "Phase manipulation", "Time shift", "Hanning filter", and "Scale factor". The "Scale factor" tab is currently selected and highlighted with a red border. Below the tabs, there is a label "Scale factor:" followed by a text input field. The input field contains the number "1". A small blue question mark icon is located in the top right corner of the main content area.

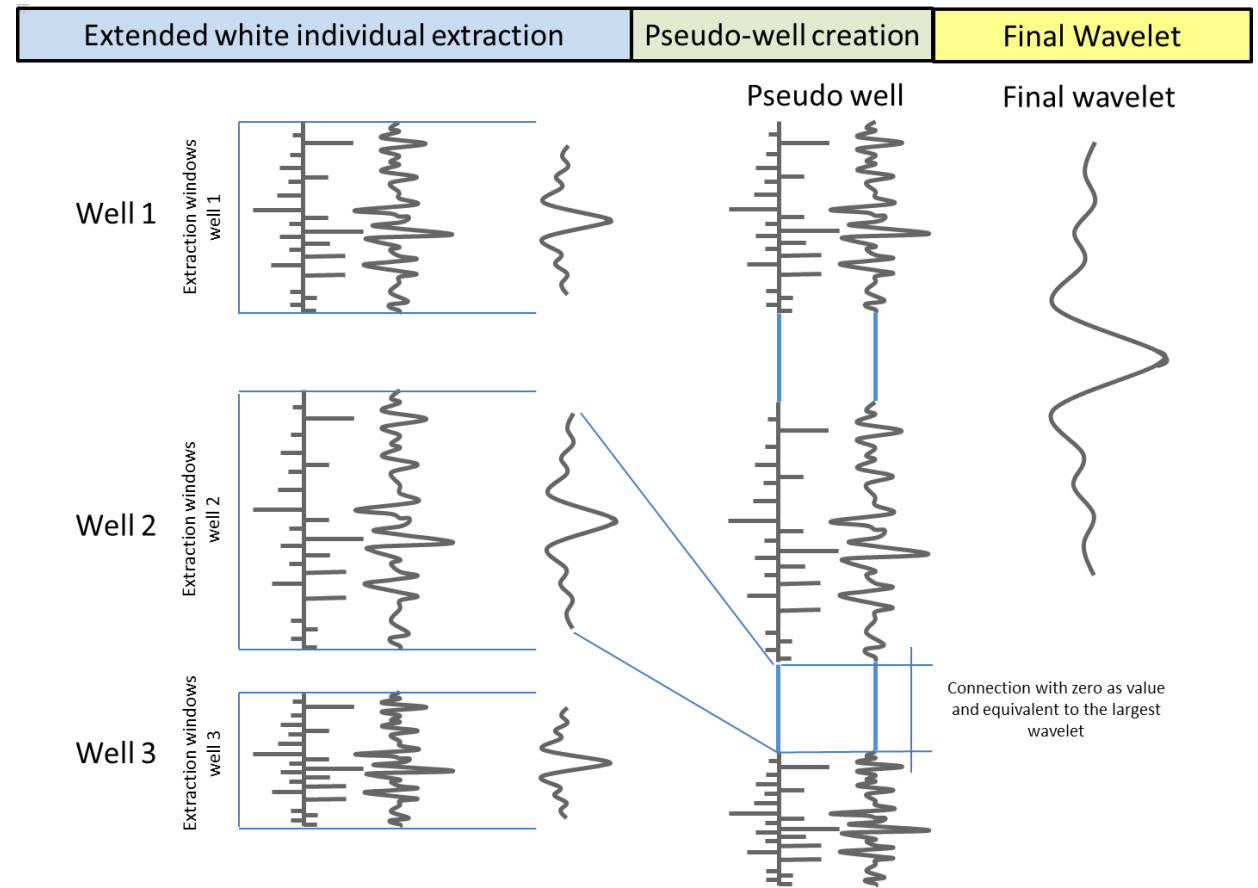
Multiwavelet

- Calculates an average wavelet from a set of selected wavelets.
- Defines the length and sample interval of the average wavelet.
- There is an additional option to invert the polarity of the resultant wavelet.



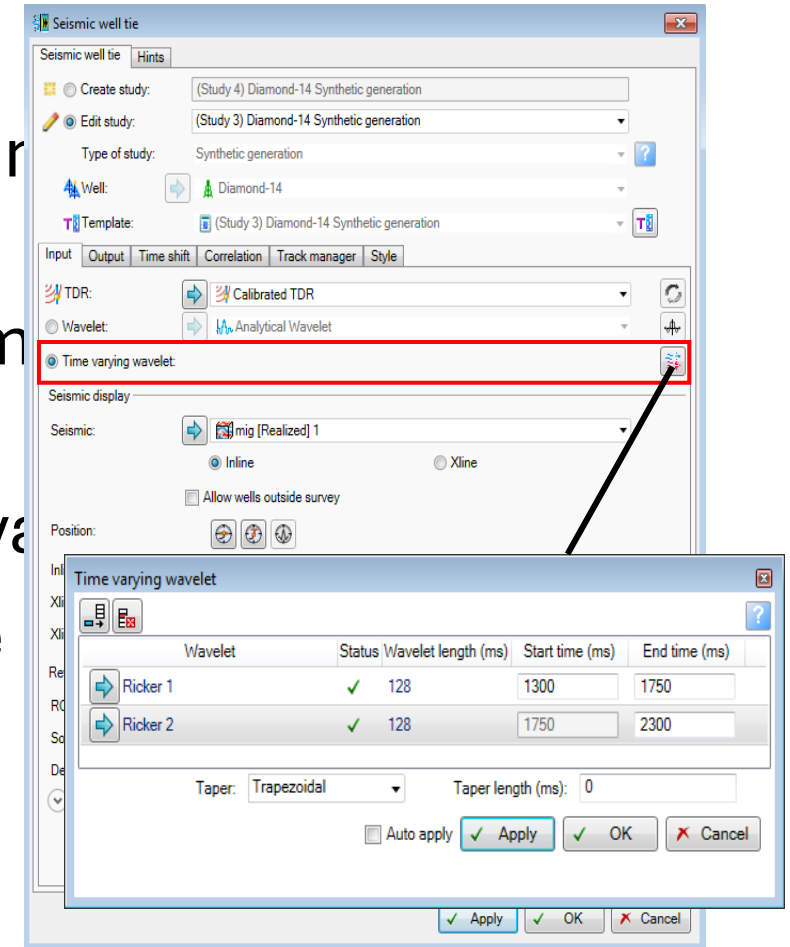
Multi-well Extended White wavelet extraction (MWEW)

- Extract a single wavelet from different wells.
- Use the extended Roy White method.
- Represent frequencies and amplitudes from different wells.



Time varying wavelet (1)

- To address decrease of frequencies and amplitudes with depth.
- Applies existing wavelets over different time intervals to generate a synthetic.
- Each wavelet is used for a specified interval.
- The minimum window that you can define



Time varying wavelet (2)

- Time ranges or time intervals can be specified over which a series of wavelets can be used to generate the synthetic.
- Create as many as ten time ranges represented in a dynamic table.

Time varying wavelet

Wavelet	Status	Wavelet length (ms)	Start time (ms)	End time (ms)
➡ Ricker 1	✓	128	1300	1750
➡ Ricker 2	✓	128	1750	2300

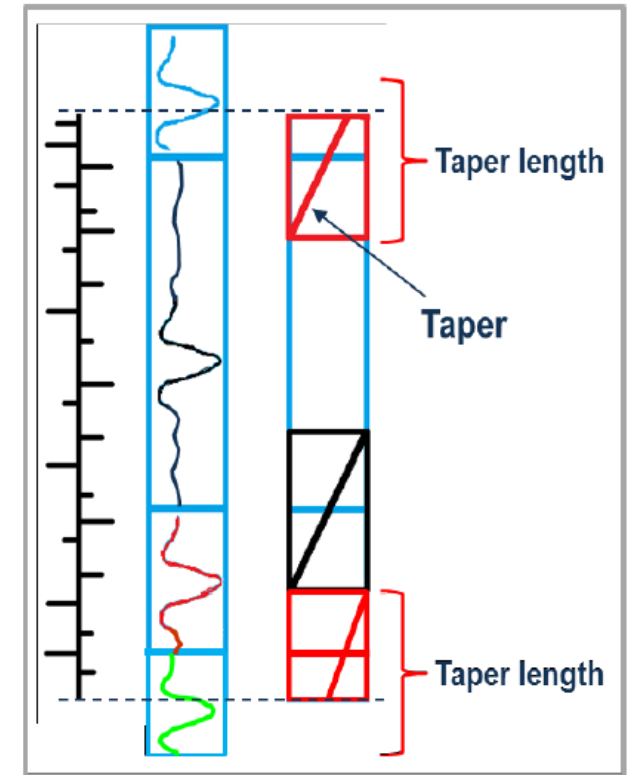
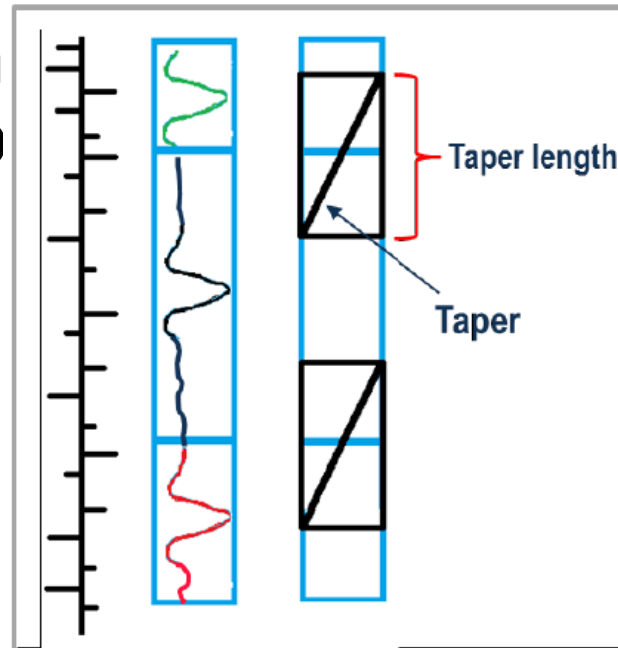
Taper: Trapezoidal Taper length (ms): 0

☐ Auto apply

Time varying wavelet (3)

Taper length:

- A time in milliseconds over which the gate is gradually smoothed to its edges.
- This parameter is an integer value with maximum limits of the smallest time range.



Time varying wavelet (4)

- All the wavelets are resampled to the same sample rate of the one with the lowest sample interval.
- A synthetic seismogram is generated using the highest resolution.

